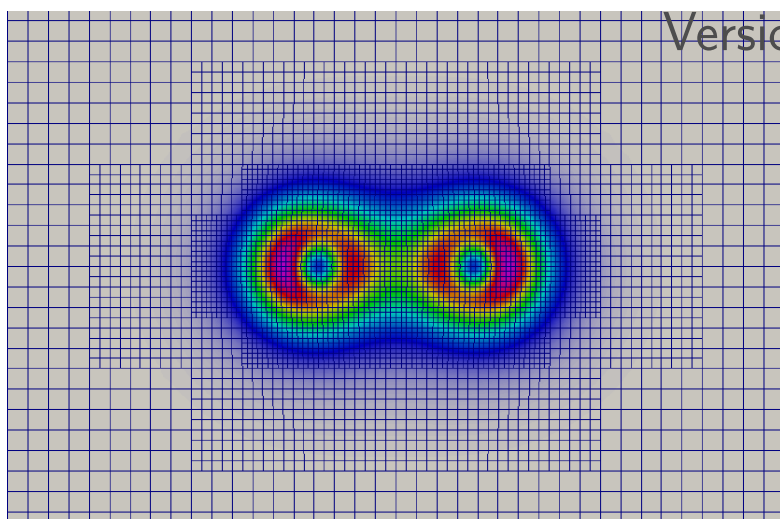


DFT-FE

Density Functional Theory calculations with Finite-Elements

User Manual

Version 1.1.0-pre (dev)
(generated June 11, 2024)



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[website of dftfe](#)

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1 Introduction

DFT-FE is a C++ code for materials modeling from first principles using Kohn-Sham density functional theory. It is based on adaptive finite-element discretization that handles all-electron and pseudopotential calculations in the same framework, and incorporates scalable and efficient solvers for the solution of the Kohn-Sham equations. Importantly, DFT-FE can handle general geometries and boundary conditions, including periodic, semi-periodic and non-periodic systems. DFT-FE code builds on top of the deal.II library for everything that has to do with finite elements, geometries, meshes, etc., and, through deal.II on p4est for parallel adaptive mesh handling.

1.1 Authors

DFT-FE is developed and maintained by the [Computational Materials Physics group at University of Michigan](#), and the [MATRIX lab, Indian Institute of Science](#). The code is maintained by a group of principal developers, who manage the architecture of the code and the core functionalities. Developers with significant contributions to core functionalities and code architecture in the past who are no longer active principal developers, are listed under principal developers emeriti. A subset of the principal developers and mentors are administrators. Finally, all contributors who have contributed to major parts of the DFT-FE code or sent important fixes and enhancements are listed under contributors. All the underlying lists are in alphabetical order.

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1.2 Acknowledgments

The development of DFT-FE open source code relating to pseudopotential calculations has been funded in part by the Department of Energy PRISMS Software Center at University of Michigan, and the Toyota Research Institute. The development of DFT-FE open source code relating to all-electron calculations has been funded by the Department of Energy Basic Energy Sciences. The methods and algorithms that have been implemented in DFT-FE are outputs from research activities over many years that have been supported by the Army Research Office, Air Force Office of Scientific Research, Department of Energy Basic Energy Sciences, National Science Foundation and Toyota Research Institute.

1.3 Referencing DFT-FE

Please refer to [referencing DFT-FE](#) to properly cite the use of DFT-FE in your scientific work.

2 Useful background information

We refer to the following articles for a background of the methods and algorithms implemented in DFT-FE.

1. P. Motamarri, S. Das, S. Rudraraju, K. Ghosh, D. Davydov, V. Gavini, DFT-FE—A massively parallel adaptive finite-element code for large-scale density functional theory calculations, *Comput. Phys. Commun.* 246, 106853 (2020).
1. P. Motamarri, M.R. Nowak, K. Leiter, J. Knap, V. Gavini, Higher-order adaptive finite-element methods for Kohn-Sham density functional theory, *J. Comput. Phys.* 253, 308-343 (2013).
3. P. Motamarri, V. Gavini, Configurational forces in electronic structure calculations using Kohn-Sham density functional theory, *Phys. Rev. B* 97 165132 (2018).

In addition, below are some useful references on finite element method and some online resources that provide a background of finite elements and their application to the solution of partial differential equations.

1. T.J.R. Hughes, The finite element method: linear static and dynamic finite element analysis, Dover Publication, 2000.
2. K.-J. Bathe, Finite element procedures, Klaus-Jürgen Bathe, 2014.
3. The finite element method for problems in physics, online course by Krishna Garikipati. [Link](#)
4. Online lectures on “Finite element methods in scientific computing” by Wolfgang Bangerth. [Link](#)

3 Installation

All the underlying installation instructions assume a Linux operating system. We assume standard tools and libraries like CMake, compilers- (C, C++ and Fortran), CUDA/HIP (in case of GPU architectures), MPI, and math(BLAS-LAPACK) libraries are pre-installed. Most high-performance computers would have the latest version of these libraries in the default environment. However, in many cases you would have to use [Environment Modules](#) to set the correct environment variables for the above and compilation tools like [CMake](#). For example, on one of the high-performance computers (UMICH Greatlakes) we develop and test the DFT-FE code, we can use the following commands to set the desired environment variables

```

$ module load cmake/3.18.2
$ module load gcc/8.2.0
$ module load openmpi/3.1.4
$ module load mkl/2018.0.4
$ module load cuda/11.0.2 (if installing for NVIDIA GPUs)
$ module load rocm/5.1.0 (if installing for AMD GPUs)

```

Note the above are shown only as an example. We strongly recommend using the latest stable version of compilers-(C, C++ and Fortran), CUDA/HIP, MPI and math libraries available on your high-performance computer. DFT-FE's installation requires also the following minimum versions of the above compilers and libraries:

- CMake 3.17.0
- GCC 8.2.0
- CUDA 11.0.2/ ROCm (if installing for NVIDIA / AMD GPUs)

We currently do not support Intel compilers due to a compilation issue of the deal.II library. Please use GNU compilers only. Further, if version of CMake greater than 3.17.0 is not available on your machine please install latest version from here [CMake](#) or use pre-installed binaries most appropriate for your machine.

3.1 Compiling and installing external libraries

DFT-FE is primarily based on the open source finite element library [deal.II](#), through which external dependencies on [p4est](#) and BLAS-LAPACK are set. The other required external libraries, which are not interfaced via deal.II are [ALGLIB](#), [Libxc](#), [spglib](#), [Libxml2](#) and [ELPA](#) (ELPA requires [ScaLAPACK](#)). DFT-FE also optionally links to [PETSc](#) (via deal.II), [SLEPc](#) (via deal.II) and to [nccl](#) (for GPU compilation). The optional dependencies of PETSc and SLEPc are only required for all-electron calculations using DFT-FE, which uses the more stable Gram-Schmidt orthogonalization routine instead of the default Cholesky-Gram-Schmidt orthogonalization. For pseudopotential calculations, PETSc and SLEPc dependencies are not required as the default Cholesky-Gram-Schmidt orthogonalization is very robust. DFT-FE can also be linked to [simple-dftd3](#) and [dftd4](#) in order to provide support for the dftd family of dispersion corrections by Grimme et. al.. Below, we give brief installation instructions for each of the above libraries.

3.1.1 Instructions for dependencies: ALGLIB, Libxc, spglib, Libxml2, ScaLAPACK, ELPA, p4est and nccl (nccl is optional)

1. **ALGLIB:** Used by DFT-FE for spline fitting for various radial data. Download the current release of the Alglib free C++ edition from <http://www.alglib.net/download.php>. After downloading and unpacking, go to `cpp/src`, and create a shared library using a C++ compiler. For example, using GCC compiler do

```

$ g++ -c -fPIC -O2 *.cpp
$ g++ *.o -shared -o libAlglib.so

```

2. **Libxc:** Used by DFT-FE for exchange-correlation functionals. Download the current release from <https://libxc.gitlab.io/download/>, and do

```

$ ./configure --prefix=libxc_install_dir_path
               CC=c_compiler CXX=c++_compiler FC=fortran_compiler
               CFLAGS="-O2 -fPIC" FCFLAGS="-O2 -fPIC" CXXFLAGS="-O2 -fPIC"

$ make
$ make install

```

Do not forget to replace `libxc_install_dir_path` by some appropriate path on your file system and make sure that you have write permissions. Also replace `c_compiler`, `c++_compiler` and `fortran_compiler` with compilers on your system.

3. **spglib**: Used by DFT-FE to find crystal symmetries. To install spglib, first obtain the development version of spglib from their github repository by

```
$ git clone https://github.com/spglib/spglib.git
```

and next follow the “Compiling using cmake” installation procedure described in <https://spglib.readthedocs.io/en/stable/install.html>. We recommend using the ccmake gui interface for the installation and also use appropriate compiler for `CMAKE_C_COMPILER`.

4. **Libxml2**: Libxml2 is used by DFT-FE to read `.xml` files. Most likely, Libxml2 might be already installed in the high-performance computer you are working with. It is usually installed in the default locations like `/usr/lib64` (library path which contains `.so` file for Libxml2, something like `libxml2.so.2`) and `/usr/include/libxml2` (include path).

Libxml2 can also be installed by doing (Do not use these instructions if you have already have Libxml2 on your system)

```
$ git clone https://gitlab.gnome.org/GNOME/libxml2.git
$ ./autogen.sh --prefix=Libxml_install_dir_path
$ make
$ make install
```

There might be errors complaining that it can not create regular file `libxml2.py` in `/usr/lib/python2.7/site-packages`, but that should not matter.

5. **ScaLAPACK**: ScaLAPACK library is used by DFT-FE via ELPA for its parallel linear algebra routines involving dense matrices, as well being a dependency for ELPA. **If Intel MKL math library is available, please skip this step, as the ScaLAPACK libraries therein can be used directly.** If Intel MKL math library is not available, Netlib ScaLAPACK <http://www.netlib.org/scalapack/> needs to be installed using the instructions below. Download the current release version (2.2.0) from http://www.netlib.org/scalapack/#_software, and build a shared library (use `BUILD_SHARED_LIBS=ON`, `BUILD_STATIC_LIBS=OFF` and `BUILD_TESTING=OFF`) installation of ScaLAPACK using cmake. We recommend using the ccmake gui interface for the installation. Further, use the appropriate compilers for `CMAKE_C_COMPILER` and `CMAKE_FORTRAN_COMPILER`, and also use `-fPIC` flag for `CMAKE_C_FLAGS` and `-fPIC -fallow-argument-mismatch` for `CMAKE_Fortran_FLAGS`. For best performance, ScaLAPACK must be linked to optimized BLAS-LAPACK libraries by using `USE_OPTIMIZED_LAPACK_BLAS=ON`, and providing external paths to BLAS-LAPACK libraries (MKL, OpenBlas, ESSL etc.) during the cmake configuration.
6. **ELPA**: ELPA library is used by DFT-FE for its parallel linear algebra routines involving dense matrices. ELPA requires the ScaLAPACK library (see above) as a dependency. Download the latest version `elpa-2024.03.001` from <https://elpa.mpcdf.mpg.de/software/> and follow the installation instructions in there. Example of ELPA installation on UMICH Greatlakes supercomputer with GNU compiler, Open MPI library, and Intel MKL math library:

```
$ cd elpaDir
$ mkdir build
$ cd build
$ ../configure --enable-openmp FC=mpif90 CC=mpicc CXX=mpicxx
FCFLAGS="-fopenmp -O2 -march=native" CFLAGS="-fopenmp -O2 -march=native"
```

```

CXXFLAGS="-fopenmp -O2 -march=native" --prefix=elpa_install_path
SCALAPACK_LDFLAGS="-L${MKLRROOT}/lib/intel64 -lmkl_scalapack_lp64
-Wl,--no-as-needed -lmkl_intel_lp64
-lmkl_gnu_thread -lmkl_core -lmkl_blacs_openmpi_lp64 -lgomp -lpthread -lm -ldl"
SCALAPACK_FCFLAGS="-L${MKLRROOT}/lib/intel64 -lmkl_scalapack_lp64
-Wl,--no-as-needed -lmkl_gf_lp64 -lmkl_gnu_thread -lmkl_core
-lmkl_blacs_openmpi_lp64 -lgomp -lpthread -lm -ldl"
$ make -j 4
$ make install

```

The MKL paths and linker flags were obtained with the help of [Intel MKL Link Line Advisor](#) for GNU Fortran compiler (Note the usage of `-lmkl_gf_lp64` flag above).

If the machine of interest has NVIDIA/AMD GPUs and CUDA/ROCm compiler, ELPA can take advantage of GPUs. For example on OLCF Summit GPU nodes, we use the following configure line

```

../configure CXX=mpic++ FC=mpif90 CC=mpicc
CXXFLAGS="-O2 -fPIC -mcpu=power9 -mvsx -maltivec"
FCFLAGS="-O2 -fPIC -mcpu=power9 -mvsx -maltivec"
CFLAGS="-O2 -fPIC -mcpu=power9 -mvsx -maltivec"
--enable-nvidia-gpu --with-NVIDIA-GPU-compute-capability="sm_70"
--enable-gpu-streams=nvidia --with-cuda-path="$OLCF_CUDA_ROOT"
--with-cuda-sdk-path="$OLCF_CUDA_ROOT"
--prefix=elpa_install_path
LDFLAGS="-L$scalapack_lib_path -lscalapack
-L$OLCF_ESSL_ROOT/lib64 -lessl
-L$OLCF_NETLIB_LAPACK_ROOT/lib64 -llapack
-L$OLCF_OPENBLAS_ROOT/lib -lopenblas"
--disable-sse --disable-sse-assembly --disable-avx
--disable-avx2 --disable-avx512 --enable-c-tests=no --enable-cpp-tests=no
--enable-option-checking=fatal --enable-shared

```

We provide another ELPA installation example on OLCF Crusher nodes with AMD GPUs and ROCm compiler:

```

../configure CXX=hipcc CC=hipcc FC=ftn
FCFLAGS="-march=znver3 -O2 -fPIC"
CFLAGS="-march=znver3 -fPIC -O2 -I${ROCM_PATH}/include
-I${ROCM_PATH}/hip/include -I${ROCM_PATH}/hipcub/include
-I${ROCM_PATH}/rocblas/include --amdgpu-target=gfx90a
-I${MPICH_DIR}/include" CXXFLAGS="-march=znver3 -std=c++14
-I${ROCM_PATH}/include -fPIC -O2 -I${ROCM_PATH}/hip/include
-I${ROCM_PATH}/hipcub/include -I${ROCM_PATH}/rocblas/include
--amdgpu-target=gfx90a -I${MPICH_DIR}/include" --enable-amd-gpu
--enable-gpu-streams=amd --prefix=elpa_install_path
LIBS="-L${ROCM_PATH}/lib -lamdhip64 -L${ROCM_PATH}/rocblas/lib
-lrocblas -L${MPICH_DIR}/lib -lmpi -L${CRAY_MPICH_ROOTDIR}/gtl/lib -lmpi_gtl_hsa
-L$scalapack_lib_path -lscalapack
-L/sw/crusher/spack-envs/base/opt/cray-sles15-zen3/gcc-11.2.0/openblas-0.3.17-fip547hyhcree2xgj5rmg
-L$netlib_lapack_lib_path -llapack"
--disable-sse --disable-sse-assembly --disable-avx --disable-avx2
--disable-avx512 --enable-c-tests=no --enable-option-checking=fatal

```

```
--enable-shared --enable-cpp-tests=no --enable-hipcub
```

Note the use of LDFLAGS instead of SCALAPACK_LDFLAGS and SCALAPACK_FCFLAGS, since we are using netlib ScaLAPACK instead of Intel MKL ScaLAPACK in the above. Also note use of

```
--disable-sse --disable-sse-assembly --disable-avx --disable-avx2 --disable-avx512
```

above. **Some or all of these options may be required for systems without Intel CPUs such as IBM Power and AMD Epyc processors depending on what they support.** Finally, we remark that to avoid errors during ELPA configuration step, please export the library paths of ScaLAPACK and blas libraries, for example:

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$blas_lib_path
```

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$netlib_lapack_lib_path
```

```
export LD_LIBRARY_PATH=
$LD_LIBRARY_PATH:$ROCM_PATH/lib:$ROCM_PATH/hip/lib:$ROCM_PATH/lib:$ROCM_PATH/roclblas/lib
```

7. **p4est**: This library is used by deal.II to create and distribute finite-element meshes across multiple processors. Download the v2.2 tarball of p4est from <https://github.com/p4est/p4est.github.io/raw/master/release/p4est-2.2.tar.gz>. Next download the p4est-setup.sh script from <https://raw.githubusercontent.com/dftfeDevelopers/dftfe/manual/p4est-setup.sh>. Use the script to automatically compile and install a debug and optimized version of p4est by doing

```
$ chmod u+x p4est-setup.sh
```

```
$ ./p4est-setup.sh p4est-x-y-z.tar.gz p4est_install_dir_path
```

8. **nccl/rccl (optional)**: nccl (CUDA) and rccl (ROCm) libraries are optional dependency for DFT-FE for optimal GPU Direct MPI collective communication calls. This library is recommended for running very large system sizes (greater than 20,000 electrons) on GPUs using DFT-FE. Caution: nccl/rccl library requires appropriate hardware support for GPU Direct MPI communication and GPU Aware MPI library. To install nccl/rccl, use the appropriate release version (corresponding to the CUDA/ROCm compiler version) from <https://github.com/NVIDIA/nccl> (CUDA) <https://github.com/ROCmSoftwarePlatform/rccl> (ROCm) and follow installation instructions therein.

3.1.2 Instructions for deal.II

Assuming the above p4est dependency is installed, we now briefly discuss the steps to compile and install deal.II library linked with the above dependencies. You need to install two variants of the deal.II library—one variant linked with real scalar type PETSc and SLEPc installations, and the other variant linked with complex scalar type PETSc and SLEPc installations.

1. Obtain the customized version of deal.II library via

```
$ git clone -b dealiiCustomizedCUDARelease https://github.com/dftfeDevelopers/dealii.git
```

2. In addition to requiring C, C++ and Fortran compilers, MPI library, and CMake, deal.II additionally requires BOOST library. If not found externally, cmake will resort to the bundled BOOST that comes along with deal.II. Based on our experience, we recommend to use the deal.II's bundled boost (enforced by unsetting/unloading external BOOST library environment paths) to avoid compilation issues. Please do not install deal.II with GPU support as that is not needed by DFT-FE. Further, when installing deal.II for (DFT-FE with GPU support), one must use `-DDEAL_II_ALLOW_PLATFORM_INTROSPECTION=OFF` to avoid compilation failure of DFT-FE.


```

3. $ mkdir build
   $ cd build
   $ cmake -DCMAKE_INSTALL_PREFIX=dealii_install_dir_path
           otherCmakeOptions ../deal.II
   $ make -j 8
   $ make install

```

“otherCmakeOptions” include the following options for CPU installation:

```

-DCMAKE_CXX_STANDARD=17
-DCMAKE_C_COMPILER=c_compiler
-DCMAKE_CXX_COMPILER=cxx_compiler
-DCMAKE_Fortran_COMPILER=fortran_compiler
-DMPI_C_COMPILER=mpi_c_compiler_wrapper
-DMPI_CXX_COMPILER=mpi_cxx_compiler_wrapper
-DMPI_Fortran_COMPILER=mpi_fortran_compiler_wrapper
-DCMAKE_CXX_FLAGS=cxx_flags
-DCMAKE_C_FLAGS=c_flags
-DDEAL_II_WITH_MPI=ON -DDEAL_II_WITH_64BIT_INDICES=ON
-DDEAL_II_WITH_P4EST=ON -DP4EST_DIR=p4est_install_dir_path
-DDEAL_II_WITH_LAPACK=ON
-DLAPACK_DIR=lapack_dir_paths (both BLAS and LAPACK directory paths)
-DLAPACK_FOUNDED=true
-DLAPACK_LIBRARIES=lapack_lib_paths (both BLAS and LAPACK library paths)
-DDEAL_II_WITH_TBB=OFF
-DDEAL_II_WITH_TASKFLOW=OFF
-DDEAL_II_COMPONENT_EXAMPLES=OFF
-DDEAL_II_FORCE_BUNDLED_BOOST=ON

```

For more information about installing deal.II library refer to <https://dealii.org/developer/readme.html>. We also provide here an example of deal.II installation, which we did on UMICH Greatlakes super-computer with GNU compiler, Open MPI library, and Intel MKL math library

```

$ mkdir build
$ cd build
$ cmake
-DCMAKE_CXX_STANDARD=17
-DCMAKE_C_COMPILER=gcc
-DCMAKE_CXX_COMPILER=g++
-DCMAKE_Fortran_COMPILER=gfortran
-DMPI_C_COMPILER=mpicc
-DMPI_CXX_COMPILER=mpicxx
-DMPI_Fortran_COMPILER=mpif90
-DCMAKE_CXX_FLAGS="-march=native -std=c++17"
-DCMAKE_C_FLAGS="-march=native -std=c++17"
-DDEAL_II_CXX_FLAGS_RELEASE="-O2"
-DDEAL_II_COMPONENT_EXAMPLES=OFF
-DDEAL_II_WITH_MPI=ON
-DDEAL_II_WITH_64BIT_INDICES=ON
-DDEAL_II_WITH_TBB=OFF
-DDEAL_II_WITH_TASKFLOW=OFF
-DDEAL_II_FORCE_BUNDLED_BOOST=ON
-DDEAL_II_WITH_P4EST=ON

```

```

-DP4EST_DIR=p4est_install_path
-DDEAL_II_WITH_LAPACK=ON -DLAPACK_DIR="${MKLRROOT}/lib/intel64"
-DLAPACK_FOUND=true
-DLAPACK_LIBRARIES="-L${MKLRROOT}/lib/intel64
-Wl,--no-as-needed -lmkl_intel_lp64
-lmkl_gnu_thread -lmkl_core -lgomp -lpthread -lm
-ldl" -DLAPACK_INCLUDE_DIRS="-I${MKLRROOT}/include"
-DCMAKE_INSTALL_PREFIX=dealii_install_path ../dealii
$ make -j 8
$ make install

```

The values for `-DLAPACK_DIR`, `-DLAPACK_LIBRARIES` and `-DLAPACK_LINKER_FLAGS` were obtained with the help of [Intel MKL Link Line Advisor](#) for GNU C++ compiler (cf. Fig. 1).

3.1.3 Instructions for simple-dftd3 and dftd4

1. **simple-dftd3**: Used by DFT-FE to provide dftd3 corrections to energy, force and stress. Download the current release (0.6.0 at the time of writing) of simple-dftd3 from <https://github.com/awvwgk/simple-dftd3/releases>. After downloading and unpacking, use cmake to build the library. For example, using GNU compiler and Intel MKL for BLAS, do

```

$ FC=gfortran cmake -B _build
-DBLAS_LIBRARIES="-L${MKLRROOT}/lib/intel64
-Wl,--no-as-needed -lmkl_gf_lp64
-lmkl_gnu_thread -lmkl_core -lgomp
-lpthread -lm -ldl"
-DCMAKE_INSTALL_PREFIX=simple_dftd3_install_path
$ cmake --build _build
$ cmake --install _build

```

2. **dftd4**: Used by DFT-FE to provide dftd4 corrections to energy, force and stress. Download the current release (3.6.0 at the time of writing) of dftd4 from <https://github.com/dftd4/dftd4/releases>. After downloading and unpacking, use cmake to build the library. For example, using GNU compiler and Intel MKL for BLAS and LAPACK, do

```

$ cd dftd4-3.6.0
$ mkdir build
$ cd build
$ cmake -DCMAKE_Fortran_COMPILER=gfortran
-DCMAKE_C_COMPILER=gcc
-DBLAS_LIBRARIES="-L${MKLRROOT}/lib/intel64 -Wl,--no-as-needed
-lmkl_intel_lp64 -lmkl_gnu_thread -lmkl_rt -lmkl_core -lgomp
-lpthread -lm -ldl"
-DLAPACK_LIBRARIES="-L${MKLRROOT}/lib/intel64 -Wl,--no-as-needed -lmkl_intel_lp64
-lmkl_gnu_thread -lmkl_rt -lmkl_core
-lgomp -lpthread -lm -ldl"
-DBUILD_SHARED_LIBS=ON
-DCMAKE_INSTALL_PREFIX=dftd4_install_path
-DWITH_OpenMP=OFF
..
$ make -j16
$ make install

```

Intel® oneAPI Math Kernel Library (oneMKL) Link Line Advisor v6.16

Reset

Select Intel® product:	oneMKL 2021
Select OS:	Linux*
Select programming language:	C/C++
Select compiler:	GNU C/C++
Select architecture:	Intel(R) 64
Select dynamic or static linking:	Dynamic
Select interface layer:	C API with 32-bit integer
Select threading layer:	OpenMP threading
Select OpenMP library:	GNU (libgomp)
Enable OpenMP offload feature to GPU:	☐
Select cluster library:	☐ Parallel Direct Sparse Solver for Clusters (BLACS required) ☐ Cluster Discrete Fast Fourier Transform (BLACS required) ☒ ScaLAPACK (BLACS required) ☒ BLACS
Select MPI library:	Intel(R) MPI
Select the Fortran 95 interfaces:	☐ BLAS95 ☐ LAPACK95
Link with Intel® oneMKL libraries explicitly:	☒
Link with DPC++ debug runtime compatible libraries:	☐
Use this link line:	
<pre>-L\$(MKLROOT)/lib/intel64 -lmkl_scalapack_lp64 -Wl,--no-as-needed -lmkl_intel_lp64 -lmkl_gnu_thread -lmkl_core -lmkl_blacs_intelmpi_lp64 -lgomp -lpthread -lm -ldl</pre>	
Compiler options:	
<pre>-m64 -I"\$(MKLROOT)/include"</pre>	
Notes:	
o Set INCLUDE, MKLROOT, TBBROOT, LD_LIBRARY_PATH, LIBRARY_PATH, CPATH and NLSPATH environment variables in the command shell using the Intel(R) oneAPI setvars script in Intel(R) oneAPI root directory. Please also see the Intel(R) oneMKL Developer Guide.	

Figure 1: Example usage of Intel MKL line advisor. Use the options in the “link line” generated by the line advisor tool. Please note to change “intelmpi” in “-lmkl_blacs_intelmpi_lp64” to “openmpi” if using openmpi MPI library.

The values for `-DBLAS_LIBRARIES` and `-DLAPACK_LIBRARIES` were obtained with the help of [Intel MKL Link Line Advisor](#) for GNU fortran compiler.

3.2 Obtaining and Compiling DFT-FE

Assuming that you have already installed the above external dependencies, next follow the steps below to obtain and compile DFT-FE.

1. Obtain the source code of the current release of DFT-FE with all current patches using [Git](#):

```
$ git clone -b release1.0 https://github.com/dftfeDevelopers/dftfe.git
$ cd dftfe
```

Do `git pull` in the `dftfe` directory any time to obtain new patches that have been added since your `git clone` or last `git pull`. If you are not familiar with Git, you may download the current release tarball from the [Downloads](#) page in our website, but downloading via Git is recommended to avail new patches seamlessly.

2. Set paths to external libraries (deal.II, ALGLIB, Libxc, spglib, Libxml2, and ELPA), C++ compiler, and C++ compiler flags in `setupUser.sh`, which is a script to compile DFT-FE using cmake. `nccl` library can also be optionally provided in case of GPU compilation. If compiling only for CPUs, set the following to OFF

```
withGPU=OFF
withDCCL=OFF
```

For GPU compilation, `withGPU`, `gpuLang`, and `gpuVendor` should be set ON, "cuda"/"hip" and "nvidia"/"amd" respectively. Also appropriate C++ compiler, device (denoting GPU) compilation and architecture flags must be passed to access CUDA/HIP compiler for device code compilation. **CAUTION! GPU enabled DFT-FE compilation must only be run on machines with GPU access.**

3. To compile DFT-FE, first create a build directory anywhere on your machine. Next from inside the build directory do

```
$ bash $dftfe_source/setupUser.sh
```

Please use the full directory path for `$dftfe_source` above. Also note that sometimes compilation on login node can crash due to insufficient memory. In those cases, we recommend using an interactive job if available on your computing cluster.

4. If compilation is successful, the following executables will be created:

```
$dftfe_build_dir/release/real/dftfe
$dftfe_build_dir/release/complex/dftfe
```

5. To compile DFT-FE in debug mode (much slower but useful for debugging), set `build_type=Debug` in `setupUser.sh` and do:

```
$ bash $dftfe_source/setupUser.h
```

which will create the following debug mode executables:

```
$dftfe_build_dir/debug/real/dftfe
$dftfe_build_dir/debug/complex/dftfe
```

3.3 Optional PETSc, SLEPc, deal.II and DFT-FE installation instructions for all-electron calculations using DFT-FE

Users can ignore this section if only interested in pseudopotential calculations. The optional dependencies of PETSc and SLEPc are only required for all-electron calculations using DFT-FE, which uses the more stable Gram-Schmidt orthogonalization routine instead of the default Cholesky-Gram-Schmidt orthogonalization.

1. **PETSc:** PETSc is a parallel linear algebra library. DFT-FE with PETSc and SLEPc dependencies needs two variants of the PETSc installation—one with real scalar type and the another with complex scalar type. Also both the installation variants must have 64-bit indices and optimized mode enabled during the installation. To install PETSc, first download the current release (3.15.0 or later) tarball from <https://www.mcs.anl.gov/petsc/download/index.html>, unpack it, and follow the installation instructions in <https://www.mcs.anl.gov/petsc/documentation/installation.html>.

Below, we show an example installation for the real scalar type variant. This example should be used only as a reference.

```
$ ./configure --prefix=petscReal_install_dir_path --with-debugging=no
--with-64-bit-indices=true --with-cc=c_compiler
--with-cxx=c++_compiler --with-fc=fortran_compiler
--with-blas-lapack-lib=(optimized BLAS-LAPACK library path)
CFLAGS=c_compiler_flags CXXFLAGS=c++_compiler_flags
FFLAGS=fortran_compiler_flags
```

```
$ make PETSC_DIR=prompted by PETSc
PETSC_ARCH=prompted by PETSc
```

```
$ make PETSC_DIR=prompted by PETSc
PETSC_ARCH=prompted by PETSc
install
```

For the complex installation variant, unpack a fresh PETSc directory (required) from the tarball and repeat the above steps with the only changes being adding `--with-scalar-type=complex` and `--with-fortran-kernels=true` to the configuration step (`./configure`) as well as providing a new installation path to `--prefix`. Below we provide example configure lines for real and complex versions on UMich Greatlakes supercomputer with GNU compiler, Open MPI library, and Intel MKL math library:

```
./configure --prefix=petsc_real_install_path --with-debugging=no
--with-64-bit-indices=true --with-cc=mpicc --with-cxx=mpicxx
--with-fc=mpif90 --with-blas-lapack-lib="-Wl,--start-group
${MKLR00T}/lib/intel64/libmkl_intel_lp64.a
${MKLR00T}/lib/intel64/libmkl_gnu_thread.a
${MKLR00T}/lib/intel64/libmkl_core.a -Wl,--end-group -lgomp -lpthread -lm -ldl"
CFLAGS="-O2" CXXFLAGS="-O2" FFLAGS="-O2"
```

```
./configure --prefix=petsc_complex_install_path --with-debugging=no
--with-64-bit-indices=true --with-cc=mpicc --with-cxx=mpicxx
--with-fc=mpif90 --with-fortran-kernels=true --with-scalar-type=complex
--with-blas-lapack-lib="-Wl,--start-group
${MKLR00T}/lib/intel64/libmkl_intel_lp64.a ${MKLR00T}/lib/intel64/libmkl_gnu_thread.a
${MKLR00T}/lib/intel64/libmkl_core.a -Wl,--end-group -lgomp -lpthread -lm -ldl"
CFLAGS="-O2" CXXFLAGS="-O2" FFLAGS="-O2"
```

2. **SLEPc:** The SLEPc library is built on top of PETSc, and it is used in DFT-FE for Gram-Schmidt Orthogonalization. To install SLEPc, first download the current release (3.15.0 or later) tarball from <http://slepc.upv.es/download/>, and then follow the installation procedure described in <http://slepc.upv.es/documentation/instal.htm>. **Important:** SLEPc installation requires PETSc to be installed first. You also need to create two separate SLEPc installations- one for PETSc installed with `--with-scalar-type=real`, and the second for PETSc installed with `--with-scalar-type=complex`.

For your reference you provide here an example installation of SLEPc for real scalar type

```
$ export PETSC_DIR=petscReal_install_dir_path
$ unset PETSC_ARCH
$ cd downloaded_slepc_dir
$ ./configure --prefix=slepcReal_install_dir_path
$ make
$ make install
```

3. **deal.II:** Assuming PETSc and SLEPc are installed, we now briefly discuss the steps to compile and install deal.II library linked with the above dependencies. You need to install two variants of the deal.II library—one variant linked with real scalar type PETSc and SLEPc installations, and the other variant linked with complex scalar type PETSc and SLEPc installations.

```
$ mkdir buildReal
$ cd buildReal
$ cmake -DCMAKE_INSTALL_PREFIX=dealii_petscReal_install_dir_path
      otherCmakeOptions ../deal.II
$ make install
```

“otherCmakeOptions” include the following options

```
-DCMAKE_C_COMPILER=c_compiler
-DCMAKE_CXX_COMPILER=cxx_compiler
-DCMAKE_Fortran_COMPILER=fortran_compiler
-DMPI_C_COMPILER=mpi_c_compiler_wrapper
-DMPI_CXX_COMPILER=mpi_cxx_compiler_wrapper
-DMPI_Fortran_COMPILER=mpi_fortran_compiler_wrapper
-DCMAKE_CXX_FLAGS=cxx_flags
-DCMAKE_C_FLAGS=c_flags
-DDEAL_II_WITH_MPI=ON -DDEAL_II_WITH_64BIT_INDICES=ON
-DDEAL_II_WITH_P4EST=ON -DP4EST_DIR=p4est_install_dir_path
-DDEAL_II_WITH_PETSC=ON -DPETSC_DIR=petscReal_install_dir_path
-DDEAL_II_WITH_SLEPC=ON -DSLEPC_DIR=slepcReal_install_dir_path
-DDEAL_II_WITH_LAPACK=ON
-DLAPACK_DIR=lapack_dir_path
-DLAPACK_FOUND=true
-DLAPACK_LIBRARIES=lapack_lib_path
-DDEAL_II_WITH_TBB=OFF
-DDEAL_II_WITH_TASKFLOW=OFF
-DDEAL_II_COMPONENT_EXAMPLES=OFF
```

4. **DFT-FE:** Follow the same instructions as in Sec. 3.2, except to modify the `setupUserPetsc.sh` script instead of the `setupUser.sh`. In `setupUserPetsc.sh`, in addition to updating the paths and flags as discussed in Sec 3.2, update the dealii installation paths from the previous step as follows:

```
dealiiPetscRealDir=dealii_petscReal_install_dir_path
dealiiPetscComplexDir=dealii_petscComplex_install_dir_path
```

3.4 Important generic instructions

- We strongly recommend to link to optimized BLAS-LAPACK library. If using Intel MKL for BLAS-LAPACK library, it is **very important** to use [Intel MKL Link Line Advisor](#) to correctly link with Intel MKL for installations of PETSc, ScaLAPACK, ELPA, deal.II, and PETSc. To exploit performance benefit from threads, we recommend (strongly recommended for the new Intel Xeon Phi (KNL) and Skylake processors) linking to threaded versions of Intel MKL libraries by using the options “threading layer” and “OpenMP library” in [Intel MKL Link Line Advisor](#).
- Use `-fPIC` compiler flag for compilation of DFT-FE and its dependencies, to prevent linking errors during DFT-FE compilation.
- **CAUTION! It is highly recommended to compile deal.II, p4est, ScaLAPACK, ELPA, DFT-FE, PETSc and SLEPc with the same compilers, same BLAS-LAPACK libraries (if applicable), and same MPI libraries. This prevents deal.II compilation issues, occurrence of run time crashes, and DFT-FE performance degradation.**

4 Running DFT-FE

After compiling DFT-FE as described in Section 3, we have now two executables — `$dftfe_build_dir/release/real/dftfe` and `$dftfe_build_dir/release/complex/dftfe`. The `$dftfe_build_dir/release/real/dftfe` executable, which uses real data-structures is sufficient for fully non-periodic problems. The executable can also be used for periodic and semi-periodic problems involving a Gamma point calculation. On the other hand the `$dftfe_build_dir/release/complex/dftfe` executable, which uses complex data-structures is required for periodic and semi-periodic problems with multiple k point sampling for Brillouin zone integration. These executables are to be used for a parallel run as follows:

```
mpirun -n N ./dftfe parameterFile.prm
```

to run with N processors.

4.1 Structuring the input file

In the above, an input file with `.prm` extension is used. This file contains input parameters as described in Section A, which can be of multiple types (`string`, `double`, `integer`, `bool` etc.). All input parameters are also conveniently indexed at the end of this manual in Section A.21. As seen in Section A, there are two types of parameters: **Global parameters** and **Parameters in section A/B/...** In **Parameters in section A/B/...**, A refers to the primary subsection name, B if present refers to a subsection inside A, and so on.

First, let's consider how to use a parameter named `PARAMETER xyz` under **Global parameters**. To set it to a value, say `value` in the `.prm` file, directly use

```
set PARAMETER xyz=value
```

Next consider a parameter named `PARAMETER xyzA` under **Parameters in section A**. To set it to a value, say `value` in the `.prm` file, use

```
subsection A
  set PARAMETER xyzA=value
end
```

Finally, consider a nested parameter named `PARAMETER xyzAB` under **Parameters in section A/B**. To set it to a value, say `value` in the `.prm` file, use

```
subsection A
  subsection B
    set PARAMETER xyzAB=value
  end
end
```

Couple of final comments— more than one parameter could be used inside the same `subsection`. For example,

```
subsection A
  set PARAMETER SUBSECTION xyzA1=value1
  set PARAMETER SUBSECTION xyzA2=value2
  subsection B
    set PARAMETER SUBSUBSECTION xyzAB1=value1
    set PARAMETER SUBSUBSECTION xyzAB2=value2
  end
end
```

Also the indentation used in the above examples is only for readability.

4.2 Demo examples

Now we will walk you through a few demo examples in the `/demo/` folder. We refer to

<https://github.com/dftfeDevelopers/dftfe-benchmarks.git>

repository for a comprehensive list of accuracy and performance examples, which are designed to cover most of the capabilities of the DFT-FE code. Further, we note that the demo examples in the `/demo/` folder do not cover all the input parameter options. To get full list of input parameters see Section A. All input parameters are also conveniently indexed at the end of this manual in Section A.21.

4.2.1 Example 1

Let us consider the first example given in the folder `/demo/ex1`, where we compute the ground state of the Nitrogen molecule using pseudopotential DFT calculations employing fully non-periodic boundary conditions. There are two input parameter files— `parameterFile_a.prm` and `parameterFile_b.prm`. `parameterFile_a.prm` is computing the ground-state and forces of the Nitrogen molecule while `parameterFile_b.prm` additionally does atomic relaxation.

Below, we provide a step by step procedure on setting up the above input parameter files, doing a total energy and force convergence study with respect to finite-element mesh discretization, and finally doing the atomic relaxation of Nitrogen molecule.

1. Set the solver mode in order to perform a single ground-state calculation

```
set SOLVER MODE = GS
```

2. The geometry of the Nitrogen molecule (N_2) system is set using input parameters under `Geometry` subsection

```
subsection Geometry
  set NATOMS=2
  set NATOM TYPES=1
  set ATOMIC COORDINATES FILE      = coordinates.inp
  set DOMAIN VECTORS FILE = domainVectors.inp
end
```


where

- NATOMS is the total number of atoms, and NATOM TYPES is the total number of atom types.
- “domainVectors.inp” (any other file name can be used), given as input to DOMAIN VECTORS FILE, is the external input file which lists the three domain vectors (in a.u) describing the 3D parallelepiped computational domain. For the current example we take a cuboidal domain with 40 a.u as the edge length. Accordingly, the “domainVectors.inp” file is formatted as

```
40.0 0.0 0.0
0.0 40.0 0.0
0.0 0.0 40.0
```

where each row corresponds to a domain vector. It is a requirement that the above vectors must form a right-handed coordinate system i.e. $(v1 \times v2) \cdot v3 > 0$.

- “coordinates.inp” (any other file name can be used), given as input to ATOMIC COORDINATES FILE, is the name of an external input file present in the same workspace which lists the Cartesian coordinates of the atoms (in a.u.) with respect to origin at the center of the domain. For this example, “coordinates.inp” is described as

```
7 5 -1.30000000E+00 0.00000000E+00 0.00000000E+00
7 5 1.30000000E+00 0.00000000E+00 0.00000000E+00
```

where each line corresponds to “atomic-charge valence-charge x y z”. Since this is a pseudopotential calculation, the valence-charge must correspond to the pseudopotential input, which we discuss in the later steps.

We require Cartesian coordinates for fully non-periodic simulation domain like above while fractional coordinates are mandatory for periodic and semi-periodic simulation domain.

3. Set the fully non-periodic boundary conditions for the problem using the subsection Boundary conditions

```
subsection Boundary conditions
  set PERIODIC1 = false
  set PERIODIC2 = false
  set PERIODIC3 = false
end
```

where PERIODIC1/2/3 sets the periodicity along the first, second, and third domain vectors. We note that DFT-FE allows for arbitrary boundary conditions.

4. Set the required DFT functional input parameters for pseudopotential calculation

```
subsection DFT functional parameters
  set EXCHANGE CORRELATION TYPE = 4
  set PSEUDOPOTENTIAL CALCULATION = true
  set PSEUDOPOTENTIAL FILE NAMES LIST = pseudo.inp
end
```

where

- The choice of “4” for EXCHANGE CORRELATION TYPE corresponds to “GGA: Perdew-Burke-Ernzerhof functional [PRL. 77, 3865 (1996)]” functional.

- “pseudo.inp”, given as input to PSEUDOPOTENTIAL FILE NAMES LIST is an external file (any other file name can be used) in the same workspace, which contains the list of pseudopotential file names in UPF format corresponding to the atom types involved in the calculations. The file is formatted as

```
7 N.upf
```

where “7” is the atomic number of Nitrogen, and N.upf is the Optimized Norm-Conserving Vanderbilt pseudopotential (ONCV) file obtained from <http://www.pseudo-dojo.org/>. **Presently, we only support Norm-Conserving pseudopotential (Troullier-Martins, ONCV) files in UPF format (version 2.0 or greater).**

5. Set the input parameters for Self-Consistent field iterative procedure.

```
subsection SCF parameters
  set MIXING PARAMETER = 0.5
  set MAXIMUM ITERATIONS      = 40
  set TEMPERATURE             = 500
  set TOLERANCE                = 5e-5
  subsection Eigen-solver parameters
    set NUMBER OF KOHN-SHAM WAVEFUNCTIONS = 12
  end
end
```

where

- “0.5” set for MIXING PARAMETER is the mixing parameter to be used in the mixing scheme.
- “40” set for MAXIMUM ITERATIONS is the maximum number of iterations allowed in SCF iterative procedure.
- “500” set for TEMPERATURE is the Fermi-Dirac smearing temperature in Kelvin.
- “5e-5” set for TOLERANCE is the SCF stopping tolerance in terms of L2 norm of the electron-density difference between two successive iterations.
- “12” set for NUMBER OF KOHN-SHAM WAVEFUNCTIONS is the Number of Kohn-Sham wavefunctions to be computed in the Eigen solve (using Chebyshev subspace iteration solve) for every SCF iteration step. This parameter is set inside the subsection Eigen-solver parameters, which is nested within SCF parameters.

6. As we are also computing the force on the atoms in this example, update the Geometry subsection in the first step to

```
subsection Geometry
  set NATOMS=2
  set NATOM TYPES=1
  set ATOMIC COORDINATES FILE      = coordinates.inp
  set DOMAIN VECTORS FILE = domainVectors.inp
  subsection Optimization
    set ION FORCE = true
  end
end
```

where the ION FORCE is set to true inside the nested subsection Optimization. This computes and prints the forces on the atoms at the end of the ground-state solve.

7. DFT-FE as mentioned before employs finite-element basis. These basis are piecewise polynomial functions. The three important FE discretization related parameters in DFT-FE, which the user needs to set are the POLYNOMIAL ORDER, MESH SIZE AROUND ATOM, and ATOM BALL RADIUS.

First, the POLYNOMIAL ORDER sets the order of the piecewise continuous FE interpolating polynomial, with higher values affording faster convergence rates with respect to discretization. Default value is set to 6. Based on our numerical investigations, we recommend POLYNOMIAL ORDER=7 for soft pseudopotentials (< 20 Ha plane-wave cutoff), POLYNOMIAL ORDER=6 for medium hard/very hard pseudopotentials, and POLYNOMIAL ORDER=5 for all-electron calculations to be most computationally efficient choices on both CPUs and GPUs.

Second, the MESH SIZE AROUND ATOM input parameter sets the size (in Bohr units) of the FE mesh element around the atoms. Mesh sizes away from the atoms and the intervening mesh adaptivity is heuristically set inside the code depending on the domain boundary conditions. The MESH SIZE AROUND ATOM is the most important parameter for the user to control in pseudopotential DFT calculations. This parameter is inversely related to the plane-wave cutoff in plane-wave based DFT codes. Smaller mesh sizes lead to lower discretization related errors, but at the cost of more degrees of freedom. Based on our numerical validation studies as will be demonstrated in many of the examples, we obtain chemical accuracy (10^{-4} Ha/atom in energy, 10^{-4} Ha/Bohr in ionic forces and 5×10^{-6} Ha/Bohr³ in cell stresses) using the following choice of discretization parameters: (POLYNOMIAL ORDER=7, MESH SIZE AROUND ATOM=1.5 — 2.5) for soft pseudopotentials, and (POLYNOMIAL ORDER=6, MESH SIZE AROUND ATOM=0.5 — 1.5) for medium hard/very hard pseudopotentials. We advice the users to use the above recommendations as starting choices only and always check the convergence of their quantity of interest with respect to MESH SIZE AROUND ATOM as described below for the current demo example. For all-electron calculations, a value of around 0.5 is a good starting choice.

Third, the ATOM BALL RADIUS input parameter denotes the radius of ball enclosing every atom (in a.u.), inside which the mesh size is set close to MESH SIZE AROUND ATOM and coarse-grained in the region outside the enclosing balls. For the default value of 0.0, a heuristically determined value is used, which is good enough for most cases but can be a bit conservative choice for fully non-periodic and semi-periodic problems as well as all-electron problems. To improve the computational efficiency user may experiment with values of ATOM BALL RADIUS ranging between 3.0 to 6.0 for pseudopotential problems, and ranging between 1.0 to 2.5 for all-electron problems.

For the current problem at hand involving N_2 molecule which is a medium hard pseudopotential we set the starting finite-element mesh parameters as below

```
subsection Finite element mesh parameters
  set POLYNOMIAL ORDER = 6
  subsection Auto mesh generation parameters
    set ATOM BALL RADIUS = 3
    set MESH SIZE AROUND ATOM = 1.6
  end
end
```

and now run the problem using the `/build/release/real/dftfe` executable on "xx" number of MPI tasks

```
mpirun -n xx ../../build/release/real/dftfe parameterFile_a.prm > outputMesh1 &
```

From the "outputMesh1" file, you can obtain information on the number of degrees of freedom in the auto-generated finite-element mesh and the ground-state energy and forces.

Repeat the above step thrice, once with `MESH SIZE AROUND ATOM = 1.6`, then with `MESH SIZE AROUND ATOM = 1.3`, and finally with `MESH SIZE AROUND ATOM = 1.0`. Now run one more time with `POLYNOMIAL ORDER = 7` while keeping `MESH SIZE AROUND ATOM = 1.0`. We recommend to run this final simulation with more MPI tasks for faster computational times. The ground-state energy per atom and force on the atomId 0 is tabulated in Table 1 for all the above cases. Upon comparing the errors in the energy and force with respect the most refined mesh (*Mesh No. 4*), we observe that for *Mesh No. 3* we obtain convergence in energy per atom to $\mathcal{O}(10^{-5})$ accuracy, and convergence in force to $\mathcal{O}(10^{-5})$ accuracy. For your reference, we have provided the output file for *Mesh No. 3* at `/demo/ex1/ex1_a.output`. DFT-FE also has the capability to write finite-element meshes with electron-density or wavefunction information to .vtu format which can be visualized using software like ParaView or Visit.

```
subsection Post-processing Options
  set WRITE DENSITY FE MESH=true
end
```

in the input parameter file, and visualizing the “densityOutput.vtu” file in ParaView.

Table 1: Nitrogen molecule ground-state energy and force convergence for demo example 1

Mesh No.	POLYNOMIAL ORDER	MESH SIZE AROUND ATOM	Total degrees of freedom per atom	Energy per atom (Hartree)	Force on atomId 0 (Hartree/Bohr)
1	6	1.6	39,595	-10.31830282	0.2947253
2	6	1.3	73,093	-10.32060716	0.2929309
3	6	1.0	141,291	-10.32084447	0.2928696
4	7	1.0	222,288	-10.32086549	0.2928277

8. Finally we discuss how to set up the input parameter file for atomic relaxation. Use the same finite-element mesh input parameters as used for *Mesh No. 3*, and update the solver mode and the input parameters in `Geometry` subsection from the fifth step to

```
set SOLVER MODE = GEOOPT
subsection Geometry
  set NATOMS=2
  set NATOM TYPES=1
  set ATOMIC COORDINATES FILE = coordinates.inp
  set DOMAIN VECTORS FILE = domainVectors.inp
  subsection Optimization
    set OPTIMIZATION MODE = ION
    set FORCE TOL = 1e-4
    set ION RELAX FLAGS FILE = relaxationFlags.inp
  end
end
```

where

- `OPTIMIZATION MODE` is set to `ION` which enables atomic relaxation.
- “1e-4” for `FORCE TOL` sets the tolerance of the maximum force (in Hartree/Bohr) on an atom when atoms are considered to be relaxed.

- “relaxationFlags.inp”, given as input to ION RELAX FLAGS FILE is an external file (any other file name can be used) in the same workspace, which specifies the permission flags (1– free to move, 0– fixed) for each coordinate axis and for all atoms. This file is described as

```
1 0 0
1 0 0
```

which marks both the Nitrogen atoms to move freely along the x axis only. Now, run the atomic relaxation problem. From the output file, you should observe that the Nitrogen molecule geometry relaxed to an equilibrium bond length of 2.0846 Bohr after 9 geometry update steps. You can also obtain the relaxed geometry from the `atomsCartCoordCurrent.chk` file generated by the code. In case of simulations with periodic or semi-periodic boundary conditions the corresponding file generated would be `atomsFracCoordCurrent.chk`. For your reference, we have provided an output file at `/demo/ex1/ex1_b.output`, which was run using 36 MPI tasks.

4.2.2 Example 2

In the previous example, we discussed how to setup and run a fully non-periodic problem. Here we briefly discuss how to setup and run the fully periodic problem (FCC Aluminium unit cell) in the folder `/demo/ex2`. There are two input parameter files– `parameterFile_a.prm` and `parameterFile_b.prm`. `parameterFile_a.prm` is for computing the ground-state and cell stress of the FCC Al unit cell, while `parameterFile_b.prm` additionally does cell stress relaxation. Important input parameters in the above parameter files, beyond what we discussed in the previous example are

1. Set the solver mode in order to perform a single ground-state calculation

```
set SOLVER MODE = GS
```

2. “coordinates.inp” given as input to ATOMIC COORDINATES FILE, is the name of an external input file present in the same workspace which lists the fractional (reduced) coordinates of the atoms. For this example, “coordinates.inp” is described as

```
13 3 0.00000000E+00 0.00000000E+00 0.00000000E+00
13 3 0.00000000E+00 0.50000000E+00 0.50000000E+00
13 3 0.50000000E+00 0.00000000E+00 0.50000000E+00
13 3 0.50000000E+00 0.50000000E+00 0.00000000E+00
```

where each line corresponds to “atomic-charge valence-charge fracx fracy fracz”. **We require fractional coordinates for fully periodic or semi-periodic simulation domains while Cartesian coordinates are mandatory for fully non-periodic simulation domain.**

3. Set fully periodic boundary conditions

```
subsection Boundary conditions
  set PERIODIC1 = true
  set PERIODIC2 = true
  set PERIODIC3 = true
end
```

4. Inside the Optimization subsection, nested within the Geometry subsection set

```
set CELL STRESS = true
```

for computing the ground state cell stress.

5. subsection Brillouin zone k point sampling options

```
set USE TIME REVERSAL SYMMETRY = true
subsection Monkhorst-Pack (MP) grid generation
  set SAMPLING POINTS 1 = 2
  set SAMPLING POINTS 2 = 2
  set SAMPLING POINTS 3 = 2
  set SAMPLING SHIFT 1 = 1
  set SAMPLING SHIFT 2 = 1
  set SAMPLING SHIFT 3 = 1
end
```

end

where

- **SAMPLING POINTS 1/2/3** sets the number of Monkhorst-Pack grid points to be used along reciprocal lattice vectors 1, 2, and 3.
- Setting **SAMPLING SHIFT 1/2/3** to 1 enables fractional shifting to be used along reciprocal lattice vectors.
- Setting **USE TIME REVERSAL SYMMETRY** to true enables use of time reversal symmetry to reduce number of k points to be solved for. For this option to work **SAMPLING SHIFT 1/2/3** must be set to 1 as done above.

6. Set

```
subsection Parallelization
  set NPKPT=2
end
```

which parallelizes the work load of the irreducible k-points across two groups of MPI tasks.

7. The same strategy for convergence of the ground state energy and force discussed in the previous example is applied to the current example to get convergence in ground state energy and cell stress. The ground-state energy per atom and hydrostatic cell stress for finite-element meshes with increasing level of refinement is tabulated in Table 2. Upon comparing the errors in the energy and force with respect the most refined mesh (*Mesh No. 3*), we observe that for *Mesh No. 1* we have obtained convergence in energy per atom to $\mathcal{O}(10^{-5})$ accuracy, and convergence in cell stress to $\mathcal{O}(10^{-7})$ accuracy. The

Table 2: FCC Al ground-state energy and hydrostatic cell-stress convergence for demo example 2

Mesh No.	POLYNOMIAL ORDER	MESH SIZE AROUND ATOM	Total degrees of freedom per atom	Energy per atom (Hartree)	Hydrostatic Cell stress (Hartree/Bohr ³)
1	5	1.6	4,394	-2.30902819	-0.000109835
2	5	1.3	7,447	-2.30899310	-0.000110529
3	6	1.3	12,663	-2.30900353	-0.000110531

output file using the mesh parameters for *Mesh No.1* is provided at `/demo/ex2/ex2_a.output` (for parameterFile_a.prm).

8. For cell stress relaxation, use parameterFile_b.prm, where we update the solver mode

```
set SOLVER MODE = GEOOPT
```

and within the **Optimization** subsection nested under **Geometry** we set

```

set STRESS TOL           = 4e-6
set OPTIMIZATION MODE    = CELL
set CELL CONSTRAINT TYPE = 1

```

where

- OPTIMIZATION MODE is set to CELL which enables cell stress relaxation.
- “4e-6” for STRESS TOL sets the tolerance of the cell stress (in a.u.) for cell stress relaxation.
- Choice of “1” for CELL CONSTRAINT TYPE enforces isotropic shape-fixed volume optimization constraint during cell stress relaxation.

For your reference, the output file for the cell stress relaxation is provided at `/demo/ex2/ex2_b.output`. From the output file, you should observe that you obtain a relaxed lattice constant of 7.699 Bohr after two geometry updates. You can also obtain the relaxed geometry from the `domainBoundingVectorsCurrent.chk` file generated by the code.

5 Finding answers to more questions

If you have questions that go beyond this manual, there are a number of resources:

- For questions about DFT-FE, installation, bugs, etc., use the [DFT-FE discussion forum](#).
- For latest news, updates, and release announcements about DFT-FE please send an email to dft-fe.admin@umich.edu, and we will add you to our announcement mailing list.
- DFT-FE is primarily based on the [deal.II library](#). If you have particular questions about deal.II, contact the mailing lists described at <https://www.dealii.org/mail.html>.
- If you have specific questions about DFT-FE that are not suitable for public and archived mailing lists, you can contact the primary developers and mentors:
 - Phani Motamarri: phanim@umich.edu.
 - Sambit Das: dsambit@umich.edu.
 - Vikram Gavini: vikramg@umich.edu (Mentor).

A Run-time input parameters

The underlying description of the input parameters also includes a “Standard/Advanced/Developer” label, which signifies whether an input parameter is a standard one, or an advanced level parameter, or a developer level one only meant for development purposes. The default values of the “Advanced” and “Developer” labelled parameters are good enough for almost all cases. However, in some cases user may need to use “Advanced” labelled parameters. For user convenience, all input parameters are also indexed at the end of this manual in Section [A.21](#).

A.1 Global parameters

- *Parameter name:* RESTART
Value: false
Default: false
Description: [Standard] If set to true solvermode triggers restart checks and modifies the input files for coordinates, domain vectors. Default: false.
Possible values: A boolean value (true or false)

- Parameter name:* RESTART FOLDER

Value: .

Default: .

Description: [Standard] Folder to store restart files.

Possible values: Any string
- Parameter name:* SOLVER MODE

Value: GS

Default: GS

Description: [Standard] DFT-FE SOLVER MODE: If GS: performs GroundState calculations. If MD: performs Molecular Dynamics Simulation. If NEB: performs a NEB calculation. If GEOOPT: performs an ion and/or cell optimization calculation. If NONE: the density is initialised with superposition of atomic densities and is written to file along with mesh data. If NSCF: The density from the restart files of the GS run are used to perform NSCF calculation at the k-points specified

Possible values: Any one of GS, MD, NEB, GEOOPT, NONE, NSCF
- Parameter name:* USE GPU

Value: false

Default: false

Description: [Standard] Use GPU for compute.

Possible values: A boolean value (true or false)
- Parameter name:* VERBOSITY

Value: 1

Default: 1

Description: [Standard] Parameter to control verbosity of terminal output. Ranges from 1 for low, 2 for medium (prints some more additional information), 3 for high (prints eigenvalues and fractional occupancies at the end of each self-consistent field iteration), and 4 for very high, which is only meant for code development purposes. VERBOSITY=0 is only used for unit testing and shouldn't be used by standard users. VERBOSITY=-1 ensures no output is printed, which is useful when DFT-FE is used as a calculator inside a larger workflow where multiple parallel DFT-FE jobs might be running, for example when using ASE or generating training data for ML workflows.

Possible values: An integer n such that $-1 \leq n \leq 5$
- Parameter name:* KEEP SCRATCH FOLDER

Value: false

Default: false

Description: [Advanced] If set to true this option does not delete the dftfeScratch folder when the dftfe object is destroyed. This is useful for debugging and code development. Default: false.

Possible values: A boolean value (true or false)
- Parameter name:* REPRODUCIBLE OUTPUT

Value: false

Default: false

Description: [Developer] Limit output to what is reproducible, i.e. don't print timing or absolute paths. This parameter is only used for testing purposes.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE STRUCTURE ENERGY FORCES DATA POST PROCESS

Value: false

Default: false

Description: [Standard] Write ground-state atomistics data to a file (structureEnergyForcesGSData*.txt) with the suffix number in the file-name denoting the geometry relaxation step number. Order: number of atoms, lattice vectors (see format for DOMAIN BOUNDING VECTORS), structure, electronic free energy, internal energy, ionic forces and finally the cell stress. Structure format is four columns with the first column being atomic number and the next three columns in fractional coordinates for periodic and semi-periodic systems and Cartesian coordinates with origin at the domain center for non-periodic systems. Ionic forces are negative of gradient of DFT free energy with respect to ionic positions with the first, second and third column in each row corresponding to the x,y and z components. Cell stress is negative of gradient of the DFT free energy with respect to affine strain components scaled by volume. Cell stress is printed as sigma_ij with i denoting the row index and j denoting the column index of the stress tensor. Atomic units are used everywhere. Default: false.

Possible values: A boolean value (true or false)

- *Parameter name:* MEM OPT MODE

Value: true

Default: true

Description: [Advanced] Uses algorithms which have lower peak memory but with a marginal performance degradation. Default: true.

Possible values: A boolean value (true or false)

A.2 Parameters in section Boundary conditions

- *Parameter name:* CONSTRAINTS FROM SERIAL DOFHANDLER

Value: false

Default: false

Description: [Developer] Create constraints from serial dofHandler.

Possible values: A boolean value (true or false)

- *Parameter name:* CONSTRAINTS PARALLEL CHECK

Value: false

Default: false

Description: [Developer] Check for consistency of constraints in parallel.

Possible values: A boolean value (true or false)

- *Parameter name:* FLOATING NUCLEAR CHARGES

Value: true

Default: true

Description: [Developer] Nuclear charges are allowed to float independent of the FEM mesh nodal positions. Only allowed for pseudopotential calculations. Internally set to false for all-electron calculations.

Possible values: A boolean value (true or false)

- Parameter name:* PERIODIC1

Value: false

Default: false

Description: [Standard] Periodicity along the first domain bounding vector.

Possible values: A boolean value (true or false)
- Parameter name:* PERIODIC2

Value: false

Default: false

Description: [Standard] Periodicity along the second domain bounding vector.

Possible values: A boolean value (true or false)
- Parameter name:* PERIODIC3

Value: false

Default: false

Description: [Standard] Periodicity along the third domain bounding vector.

Possible values: A boolean value (true or false)
- Parameter name:* POINT WISE DIRICHLET CONSTRAINT

Value: false

Default: false

Description: [Developer] Flag to set point wise dirichlet constraints to eliminate null-space associated with the discretized Poisson operator subject to periodic BCs.

Possible values: A boolean value (true or false)
- Parameter name:* MULTIPOLE BOUNDARY CONDITIONS

Value: false

Default: false

Description: [Standard] Flag to set point wise multipole boundary conditions (upto quadrupole term) for non-periodic systems.

Possible values: A boolean value (true or false)
- Parameter name:* SELF POTENTIAL RADIUS

Value: 0.0

Default: 0.0

Description: [Advanced] The radius (in a.u) of the ball around an atom in which self-potential of the associated nuclear charge is solved. For the default value of 0.0, the radius value is automatically determined to accommodate the largest radius possible for the given finite element mesh. The default approach works for most problems.

Possible values: A floating point number v such that $0 \leq v \leq 50$
- Parameter name:* SMEARED NUCLEAR CHARGES

Value: true

Default: true

Description: [Developer] Nuclear charges are smeared for solving electrostatic fields. Default is true for pseudopotential calculations and false for all-electron calculations.

Possible values: A boolean value (true or false)

A.3 Parameters in section Brillouin zone k point sampling options

- *Parameter name:* USE GROUP SYMMETRY
Value: false
Default: false
Description: [Standard] Flag to control the use of point group symmetries. Currently this feature cannot be used if ION FORCE or CELL STRESS input parameters are set to true.
Possible values: A boolean value (true or false)
- *Parameter name:* USE TIME REVERSAL SYMMETRY
Value: false
Default: false
Description: [Standard] Flag to control the use of time reversal symmetry.
Possible values: A boolean value (true or false)
- *Parameter name:* kPOINT RULE FILE
Value:
Default:
Description: [Developer] File providing list of k points on which eigen values are to be computed from converged KS Hamiltonian. The first three columns specify the crystal coordinates of the k points. The fourth column provides weights of the corresponding points, which is currently not used. The eigen values are written on an output file bands.out
Possible values: Any string

A.4 Parameters in section Brillouin zone k point sampling options/Monkhorst-Pack (MP) grid generation

- *Parameter name:* SAMPLING POINTS 1
Value: 1
Default: 1
Description: [Standard] Number of Monkhorst-Pack grid points to be used along reciprocal lattice vector 1.
Possible values: An integer n such that $1 \leq n \leq 1000$
- *Parameter name:* SAMPLING POINTS 2
Value: 1
Default: 1
Description: [Standard] Number of Monkhorst-Pack grid points to be used along reciprocal lattice vector 2.
Possible values: An integer n such that $1 \leq n \leq 1000$
- *Parameter name:* SAMPLING POINTS 3
Value: 1
Default: 1
Description: [Standard] Number of Monkhorst-Pack grid points to be used along reciprocal lattice vector 3.
Possible values: An integer n such that $1 \leq n \leq 1000$

- *Parameter name:* SAMPLING SHIFT 1
Value: 0
Default: 0
Description: [Standard] If fractional shifting to be used (0 for no shift, 1 for shift) along reciprocal lattice vector 1.
Possible values: An integer n such that $0 \leq n \leq 1$
- *Parameter name:* SAMPLING SHIFT 2
Value: 0
Default: 0
Description: [Standard] If fractional shifting to be used (0 for no shift, 1 for shift) along reciprocal lattice vector 2.
Possible values: An integer n such that $0 \leq n \leq 1$
- *Parameter name:* SAMPLING SHIFT 3
Value: 0
Default: 0
Description: [Standard] If fractional shifting to be used (0 for no shift, 1 for shift) along reciprocal lattice vector 3.
Possible values: An integer n such that $0 \leq n \leq 1$

A.5 Parameters in section DFT functional parameters

- *Parameter name:* EXCHANGE CORRELATION TYPE
Value: 1
Default: 1
Description: [Standard] Parameter specifying the type of exchange-correlation to be used: 1(LDA: Perdew Zunger Ceperley Alder correlation with Slater Exchange[PRB. 23, 5048 (1981)]), 2(LDA: Perdew-Wang 92 functional with Slater Exchange [PRB. 45, 13244 (1992)]), 3(LDA: Vosko, Wilk & Nusair with Slater Exchange[Can. J. Phys. 58, 1200 (1980)]), 4(GGA: Perdew-Burke-Ernzerhof functional [PRL. 77, 3865 (1996)]), 5(RPBE: B. Hammer, L. B. Hansen, and J. K. Nørskov, Phys. Rev. B 59, 7413 (1999)), 6(ML-XC>NNLDA: LDA-PW + NN), 7(ML-XC>NNGGA: GGA-PBE + NN). Caution: options 6-7 are experimental and only accessible to the DFT-FE developers currently.
Possible values: An integer n such that $1 \leq n \leq 8$
- *Parameter name:* MODEL XC INPUT FILE
Value:
Default:
Description: [Developer] File that contains both the pytorch ML-XC NN model (.ptc file) and the tolerances. This is an experimental feature to test out any new XC functional developed using machine learning.
Possible values: Any string
- *Parameter name:* NET CHARGE
Value: 0
Default: 0

Description: [Standard] Net charge of the system in atomic units, positive quantity implies addition of electrons. In case of non-periodic boundary conditions, this capability is implemented using multipole Dirichlet inhomogeneous boundary conditions for the electrostatics. In case of periodic and semi-periodic conditions a uniform background charge is used to create a neutral system.

Possible values: A floating point number v such that $-\text{MAX_DOUBLE} \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* PSEUDOPOTENTIAL CALCULATION

Value: true

Default: true

Description: [Standard] Boolean Parameter specifying whether pseudopotential DFT calculation needs to be performed. For all-electron DFT calculation set to false.

Possible values: A boolean value (true or false)

- *Parameter name:* PSEUDOPOTENTIAL FILE NAMES LIST

Value: pseudo.inp

Default:

Description: [Standard] Pseudopotential file. This file contains the list of pseudopotential file names in UPF format corresponding to the atoms involved in the calculations. UPF version 2.0 or greater and norm-conserving pseudopotentials(ONCV and Troullier Martins) in UPF format are only accepted. File format (example for two atoms Mg(z=12), Al(z=13)): 12 filename1.upf(row1), 13 filename2.upf(row2). Important Note: ONCV pseudopotentials data base in UPF format can be downloaded from http://www.quantum-simulation.org/potentials/sg15_oncv or <http://www.pseudo-dojo.org/>. Troullier-Martins pseudopotentials in UPF format can be downloaded from <http://www.quantum-espresso.org/pseudopotentials/pp-from-abinit-web-site>.

Possible values: Any string

- *Parameter name:* PSEUDO TESTS FLAG

Value: false

Default: false

Description: [Developer] Boolean parameter specifying the explicit path of pseudopotential upf format files used for ctests

Possible values: A boolean value (true or false)

- *Parameter name:* PSP CUTOFF IMAGE CHARGES

Value: 15.0

Default: 15.0

Description: [Standard] Distance from the domain till which periodic images will be considered for the local part of the pseudopotential. Units in a.u.

Possible values: A floating point number v such that $-\text{MAX_DOUBLE} \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* SPIN POLARIZATION

Value: 0

Default: 0

Description: [Standard] Spin polarization: 0 for no spin polarization and 1 for collinear spin polarization calculation. Default option is 0.

Possible values: An integer n such that $0 \leq n \leq 1$

- *Parameter name:* START MAGNETIZATION

Value: 0.0

Default: 0.0

Description: [Standard] Starting magnetization to be used for spin-polarized DFT calculations (must be between -0.5 and +0.5). Corresponding magnetization per simulation domain will be (2 x START MAGNETIZATION x Number of electrons) a.u.

Possible values: A floating point number v such that $-0.5 \leq v \leq 0.5$

A.6 Parameters in section DFT functional parameters/Dispersion Correction

- *Parameter name:* CN CUTOFF

Value: 40.0

Default: 40.0

Description: [Advanced] Cutoff in a.u. for computing coordination number in D3 correction

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* D3 ATM

Value: false

Default: false

Description: [Standard] Boolean parameter specifying whether or not the triple dipole correction in DFTD3 is to be included (ignored if DAMPING PARAMETERS FILE is specified).

Possible values: A boolean value (true or false)

- *Parameter name:* D3 DAMPING TYPE

Value: 3

Default: 3

Description: [Standard] The damping used for DFTD3, 0 for zero damping, 1 for BJ damping, 2 for D3M variant, 3 for BJM variant (default) and 4 for the OP variant.

Possible values: An integer n such that $0 \leq n \leq 4$

- *Parameter name:* D4 MBD

Value: false

Default: false

Description: [Standard] Boolean parameter specifying whether or not the MBD correction in DFTD4 is to be included (ignored if DAMPING PARAMETERS FILE is specified).

Possible values: A boolean value (true or false)

- *Parameter name:* DAMPING PARAMETERS FILE

Value:

Default:

Description: [Advanced] Name of the file containing custom damping parameters, for ZERO damping 6 parameters are expected (s6, s8, s9, sr6, sr8, alpha), for BJ and BJM damping 6 parameters are expected (s6, s8, s9, a1, a2, alpha), for ZEROM damping 7 parameters are expected (s6, s8, s9, sr6, sr8, alpha, beta) and for optimized power damping 7 parameters are expected (s6, s8, s9, a1, a2, alpha, beta).

Possible values: Any string

- *Parameter name:* DISPERSION CORRECTION TYPE
Value: 0
Default: 0
Description: [Standard] The dispersion correction type to be included post scf convergence: 0 for none, 1 for DFT-D3[JCP 132, 154104 (2010)][JCC 32, 1456 (2011)], 2 for DFT-D4 [JCP 147, 034112 (2017)][JCP 150, 154122 (2019)][PCCP 22, 8499-8512 (2020)].
Possible values: An integer n such that $0 \leq n \leq 2$
- *Parameter name:* THREE BODY CUTOFF
Value: 40.0
Default: 40.0
Description: [Advanced] Cutoff in a.u. for computing 3 body interactions terms in D3 correction
Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$
- *Parameter name:* TWO BODY CUTOFF
Value: 94.8683298050514
Default: 94.8683298050514
Description: [Advanced] Cutoff in a.u. for computing 2 body interactions terms in D3 correction
Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

A.7 Parameters in section Finite element mesh parameters

- *Parameter name:* POLYNOMIAL ORDER
Value: 6
Default: 6
Description: [Standard] The degree of the finite-element interpolating polynomial in the Kohn-Sham Hamiltonian except the electrostatics. Default value is 6 which is good choice for most pseudopotential calculations. POLYNOMIAL ORDER= 4 or 5 is usually a good choice for all-electron problems.
Possible values: An integer n such that $1 \leq n \leq 12$
- *Parameter name:* POLYNOMIAL ORDER ELECTROSTATICS
Value: 0
Default: 0
Description: [Standard] The degree of the finite-element interpolating polynomial for the electrostatics part of the Kohn-Sham Hamiltonian. It is automatically set to POLYNOMIAL ORDER if POLYNOMIAL ORDER ELECTROSTATICS set to default value of zero.
Possible values: An integer n such that $0 \leq n \leq 24$

A.8 Parameters in section Finite element mesh parameters/Auto mesh generation parameters

- *Parameter name:* ATOM BALL RADIUS
Value: 0.0
Default: 0.0
Description: [Standard] Radius of ball enclosing every atom, inside which the mesh size is set close to MESH SIZE AROUND ATOM and coarse-grained in the region outside the enclosing balls. For the

default value of 0.0, a heuristically determined value is used, which is good enough for most cases but can be a bit conservative choice for fully non-periodic and semi-periodic problems as well as all-electron problems. To improve the computational efficiency user may experiment with values of ATOM BALL RADIUS ranging between 3.0 to 6.0 for pseudopotential problems, and ranging between 1.0 to 2.5 for all-electron problems. Units: a.u.

Possible values: A floating point number v such that $0 \leq v \leq 20$

- *Parameter name:* AUTO ADAPT BASE MESH SIZE

Value: true

Default: true

Description: [Developer] Automatically adapt the BASE MESH SIZE such that subdivisions of that during refinement leads closest to the desired MESH SIZE AROUND ATOM. Default: true.

Possible values: A boolean value (true or false)

- *Parameter name:* BASE MESH SIZE

Value: 0.0

Default: 0.0

Description: [Advanced] Mesh size of the base mesh on which refinement is performed. For the default value of 0.0, a heuristically determined base mesh size is used, which is good enough for most cases. Standard users do not need to tune this parameter. Units: a.u.

Possible values: A floating point number v such that $0 \leq v \leq 20$

- *Parameter name:* ERROR ESTIMATE WAVEFUNCTIONS

Value: 5

Default: 5

Description: [Developer] Number of wavefunctions to be used for error estimation.

Possible values: An integer n such that $0 \leq n \leq 2147483647$

- *Parameter name:* GAUSSIAN CONSTANT FORCE GENERATOR

Value: 0.75

Default: 0.75

Description: [Developer] Force computation generator gaussian constant. Also used for mesh movement. $\Gamma(r) = \exp(-(r/\text{gaussianConstant});(\text{gaussianOrder}))$.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* GAUSSIAN ORDER FORCE GENERATOR

Value: 4.0

Default: 4.0

Description: [Developer] Force computation generator gaussian order. Also used for mesh movement. $\Gamma(r) = \exp(-(r/\text{gaussianConstant});(\text{gaussianOrder}))$.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* GAUSSIAN ORDER MOVE MESH TO ATOMS

Value: 4.0

Default: 4.0

Description: [Developer] Move mesh to atoms gaussian order. $\Gamma(r) = \exp(-(r/\text{gaussianConstant});(\text{gaussianOrder}))$.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- Parameter name:* INNER ATOM BALL RADIUS

Value: 0.0

Default: 0.0

Description: [Advanced] Radius of ball enclosing every atom, inside which the mesh size is set close to MESH SIZE AT ATOM. Standard users do not need to tune this parameter. Units: a.u.

Possible values: A floating point number v such that $0 \leq v \leq 20$
- Parameter name:* MESH ADAPTION

Value: false

Default: false

Description: [Developer] Generates adaptive mesh based on a-posteriori mesh adaption strategy using single atom wavefunctions before computing the ground-state. Default: false.

Possible values: A boolean value (true or false)
- Parameter name:* MESH SIZE AROUND ATOM

Value: 1.0

Default: 1.0

Description: [Standard] Mesh size in a ball of radius ATOM BALL RADIUS around every atom. For pseudopotential calculations, the value ranges between 0.8 to 2.5 depending on the cutoff energy for the pseudopotential. For all-electron calculations, a value of around 0.5 would be a good starting choice. In most cases, MESH SIZE AROUND ATOM is the only parameter to be tuned to achieve the desired accuracy in energy and forces with respect to the mesh refinement. Units: a.u.

Possible values: A floating point number v such that $0.0001 \leq v \leq 10$
- Parameter name:* MESH SIZE AT ATOM

Value: 0.0

Default: 0.0

Description: [Advanced] Mesh size of the finite elements in the immediate vicinity of the atom. For the default value of 0.0, a heuristically determined MESH SIZE AT ATOM is used for all-electron calculations. For pseudopotential calculations, the default value of 0.0, sets the MESH SIZE AT ATOM to be the same value as MESH SIZE AROUND ATOM. Standard users do not need to tune this parameter. Units: a.u.

Possible values: A floating point number v such that $0 \leq v \leq 10$
- Parameter name:* NUM LEVELS

Value: 10

Default: 10

Description: [Developer] Number of times to be refined.

Possible values: An integer n such that $0 \leq n \leq 30$
- Parameter name:* TOLERANCE FOR MESH ADAPTION

Value: 1

Default: 1

Description: [Developer] Tolerance criteria used for stopping the multi-level mesh adaption done a-priori using single atom wavefunctions. This is used as Kinetic energy change between two successive iterations

Possible values: A floating point number v such that $0 \leq v \leq 1$

- *Parameter name:* TOP FRAC
Value: 0.1
Default: 0.1
Description: [Developer] Top fraction of elements to be refined.
Possible values: A floating point number v such that $0 \leq v \leq 1$
- *Parameter name:* USE FLAT TOP GENERATOR
Value: false
Default: false
Description: [Developer] Use a composite generator flat top and Gaussian generator for mesh movement and configurational force computation.
Possible values: A boolean value (true or false)
- *Parameter name:* USE MESH SIZES FROM ATOM LOCATIONS FILE
Value: false
Default: false
Description: [Developer] Use mesh sizes from atom locations file.
Possible values: A boolean value (true or false)

A.9 Parameters in section GPU

- *Parameter name:* AUTO GPU BLOCK SIZES
Value: true
Default: true
Description: [Advanced] Automatically sets total number of kohn-sham wave functions and eigensolver optimal block sizes for running on GPUs. If manual tuning is desired set this parameter to false and set the block sizes using the input parameters for the block sizes. Default: true.
Possible values: A boolean value (true or false)
- *Parameter name:* FINE GRAINED GPU TIMINGS
Value: false
Default: false
Description: [Developer] Print more fine grained GPU timing results. Default: false.
Possible values: A boolean value (true or false)
- *Parameter name:* SUBSPACE ROT FULL CPU MEM
Value: true
Default: true
Description: [Developer] Option to use full NxN memory on CPU in subspace rotation and when mixed precision optimization is not being used. This reduces the number of MPI_Allreduce communication calls. Default: true.
Possible values: A boolean value (true or false)

- *Parameter name:* USE DCCL
Value: false
Default: false
Description: [Advanced] Use NCCL/RCCL for GPUDIRECT communications. Default: false.
Possible values: A boolean value (true or false)
- *Parameter name:* USE ELPA GPU KERNEL
Value: false
Default: false
Description: [Advanced] If DFT-FE is linked to ELPA eigensolver library configured to run on GPUs, this parameter toggles the use of ELPA GPU kernels for dense symmetric matrix diagonalization calls in DFT-FE. ELPA version $\geq 2020.11.001$ is required for this feature. Default: false.
Possible values: A boolean value (true or false)
- *Parameter name:* USE GPUDIRECT MPI ALL REDUCE
Value: false
Default: false
Description: [Advanced] Use GPUDIRECT MPI_Allreduce. This route will only work if DFT-FE is either compiled with NVIDIA NCCL library or with GPUAwareMPI=ON. Both these routes require GPU Aware MPI library to be available as well relevant hardware. If both NVIDIA NCCL library and with GPUAwareMPI modes are toggled on, the NCCL mode takes precedence. Also note that one MPI rank per GPU can be used when using this option. Default: false.
Possible values: A boolean value (true or false)

A.10 Parameters in section Geometry

- *Parameter name:* ATOMIC COORDINATES FILE
Value: coordinates.inp
Default:
Description: [Standard] Atomic-coordinates input file name. For fully non-periodic domain give Cartesian coordinates of the atoms (in a.u) with respect to origin at the center of the domain. For periodic and semi-periodic domain give fractional coordinates of atoms. File format (example for two atoms): Atom1-atomic-charge Atom1-valence-charge x1 y1 z1 (row1), Atom2-atomic-charge Atom2-valence-charge x2 y2 z2 (row2). The number of rows must be equal to NATOMS, and number of unique atoms must be equal to NATOM TYPES.
Possible values: Any string
- *Parameter name:* ATOMIC DISP COORDINATES FILE
Value:
Default:
Description: [Standard] Atomic displacement coordinates input file name. The FEM mesh is deformed using Gaussian functions attached to the atoms. File format (example for two atoms): delx1 dely1 delz1 (row1), delx2 dely2 delz2 (row2). The number of rows must be equal to NATOMS. Units in a.u.
Possible values: Any string

- *Parameter name:* DOMAIN VECTORS FILE

Value: domainBoundingVectors.inp

Default:

Description: [Standard] Domain vectors input file name. Domain vectors are the vectors bounding the three edges of the 3D parallelepiped computational domain. File format: v1x v1y v1z (row1), v2x v2y v2z (row2), v3x v3y v3z (row3). Units: a.u. CAUTION: please ensure that the domain vectors form a right-handed coordinate system i.e. $\text{dotProduct}(\text{crossProduct}(\mathbf{v1}, \mathbf{v2}), \mathbf{v3}) > 0$. Domain vectors are the typical lattice vectors in a fully periodic calculation.

Possible values: Any string

- *Parameter name:* NATOMS

Value: 1

Default: 0

Description: [Standard] Total number of atoms. This parameter requires a mandatory non-zero input which is equal to the number of rows in the file passed to ATOMIC COORDINATES FILE.

Possible values: An integer n such that $0 \leq n \leq 2147483647$

- *Parameter name:* NATOM TYPES

Value: 1

Default: 0

Description: [Standard] Total number of atom types. This parameter requires a mandatory non-zero input which is equal to the number of unique atom types in the file passed to ATOMIC COORDINATES FILE.

Possible values: An integer n such that $0 \leq n \leq 2147483647$

A.11 Parameters in section Geometry/Optimization

- *Parameter name:* BFGS STEP METHOD

Value: QN

Default: QN

Description: [Standard] Method for computing update step in BFGS. Quasi-Newton step (default) or Rational Function Step as described in JPC 1985, 89:52-57.

Possible values: Any one of QN, RFO

- *Parameter name:* CELL CONSTRAINT TYPE

Value: 12

Default: 12

Description: [Standard] Cell relaxation constraint type, 1 (isotropic shape-fixed volume optimization), 2 (volume-fixed shape optimization), 3 (relax along domain vector component v1x), 4 (relax along domain vector component v2y), 5 (relax along domain vector component v3z), 6 (relax along domain vector components v2y and v3z), 7 (relax along domain vector components v1x and v3z), 8 (relax along domain vector components v1x and v2y), 9 (relax along domain vector components v1x, v2y and v3z), 10 (2D - relax along x and y components), 11(2D- relax only x and y components with inplane area fixed), 12(relax all domain vector components), 13 automatically decides the constraints based on boundary conditions. CAUTION: A majority of these options only make sense in an orthorhombic cell geometry.

Possible values: An integer n such that $1 \leq n \leq 13$

- Parameter name:* CELL OPT SOLVER

Value: LBFGS

Default: LBFGS

Description: [Standard] Method for Cell relaxation solver. LBFGS is the default

Possible values: Any one of BFGS, LBFGS, CGPRP
- Parameter name:* CELL STRESS

Value: false

Default: false

Description: [Standard] Boolean parameter specifying if cell stress needs to be computed. Automatically set to true if CELL OPT is true.

Possible values: A boolean value (true or false)
- Parameter name:* FORCE TOL

Value: 1e-4

Default: 1e-4

Description: [Standard] Sets the tolerance on the maximum force (in a.u.) on an atom during atomic relaxation, when the atoms are considered to be relaxed.

Possible values: A floating point number v such that $0 \leq v \leq 1$
- Parameter name:* ION FORCE

Value: false

Default: false

Description: [Standard] Boolean parameter specifying if atomic forces are to be computed. Automatically set to true if ION OPT is true.

Possible values: A boolean value (true or false)
- Parameter name:* ION OPT SOLVER

Value: LBFGS

Default: LBFGS

Description: [Standard] Method for Ion relaxation solver. LBFGS is the default

Possible values: Any one of BFGS, LBFGS, CGPRP
- Parameter name:* ION RELAX FLAGS FILE

Value:

Default:

Description: [Standard] File specifying the permission flags (1-free to move, 0-fixed) and external forces for the 3-coordinate directions and for all atoms. File format (example for two atoms with atom 1 fixed and atom 2 free and 0.01 Ha/Bohr force acting on atom 2): 0 0 0 0.0 0.0 0.0(row1), 1 1 1 0.0 0.0 0.0(row2). External forces are optional.

Possible values: Any string
- Parameter name:* LBFGS HISTORY

Value: 5

Default: 5

Description: [Standard] Number of previous steps to considered for the LBFGS update.

Possible values: An integer n such that $1 \leq n \leq 20$

- Parameter name:* MAXIMUM CELL UPDATE STEP

Value: 0.1

Default: 0.1

Description: [Standard] Sets the maximum allowed step size (deformation) during cell relaxation.

Possible values: A floating point number v such that $0 \leq v \leq 5$
- Parameter name:* MAXIMUM ION UPDATE STEP

Value: 0.5

Default: 0.5

Description: [Standard] Sets the maximum allowed step size (displacement in a.u.) during ion relaxation.

Possible values: A floating point number v such that $0 \leq v \leq 5$
- Parameter name:* MAXIMUM OPTIMIZATION STEPS

Value: 300

Default: 300

Description: [Standard] Sets the maximum number of optimization steps to be performed.

Possible values: An integer n such that $1 \leq n \leq 1000$
- Parameter name:* MAXIMUM STAGGERED CYCLES

Value: 300

Default: 300

Description: [Standard] Sets the maximum number of staggered ion/cell optimization cycles to be performed.

Possible values: An integer n such that $1 \leq n \leq 1000$
- Parameter name:* MAX LINE SEARCH ITER

Value: 5

Default: 5

Description: [Standard] Sets the maximum number of line search iterations in the case of CGPRP. Default is 5.

Possible values: An integer n such that $1 \leq n \leq 100$
- Parameter name:* NON SELF CONSISTENT FORCE

Value: false

Default: false

Description: [Developer] Boolean parameter specifying whether to include the force contributions arising out of non self-consistency in the Kohn-Sham ground-state calculation. Currently non self-consistent force computation is still in experimental phase. The default option is false.

Possible values: A boolean value (true or false)
- Parameter name:* OPTIMIZATION MODE

Value: ION

Default: ION

Description: [Standard] Specifies whether the ionic coordinates and/or the lattice vectors are relaxed.

Possible values: Any one of ION, CELL, IONCELL

- *Parameter name:* REUSE DENSITY
Value: 1
Default: 1
Description: [Standard] Parameter controlling the reuse of ground-state density during geometry optimization. The options are 0 (reinitialize density based on superposition of atomic densities), 1 (reuse ground-state density of previous relaxation step), and 2 (subtract superposition of atomic densities from the previous step's ground-state density and add superposition of atomic densities from the new atomic positions. Option 2 is not enabled for spin-polarized case. Default setting is 0.
Possible values: An integer n such that $0 \leq n \leq 2$
- *Parameter name:* REUSE WFC
Value: true
Default: true
Description: [Standard] Reuse previous ground-state wavefunctions during geometry optimization. Default setting is true.
Possible values: A boolean value (true or false)
- *Parameter name:* STRESS TOL
Value: 1e-6
Default: 1e-6
Description: [Standard] Sets the tolerance of the cell stress (in a.u.) during cell-relaxation.
Possible values: A floating point number v such that $0 \leq v \leq 1$
- *Parameter name:* USE PRECONDITIONER
Value: false
Default: false
Description: [Standard] Boolean parameter specifying if the preconditioner described by JCP 144, 164109 (2016) is to be used.
Possible values: A boolean value (true or false)

A.12 Parameters in section Helmholtz problem parameters

- *Parameter name:* ABSOLUTE TOLERANCE HELMHOLTZ
Value: 1e-10
Default: 1e-10
Description: [Advanced] Absolute tolerance on the residual as stopping criterion for Helmholtz problem convergence.
Possible values: A floating point number v such that $0 \leq v \leq 1$
- *Parameter name:* MAXIMUM ITERATIONS HELMHOLTZ
Value: 10000
Default: 10000
Description: [Advanced] Maximum number of iterations to be allowed for Helmholtz problem convergence.
Possible values: An integer n such that $0 \leq n \leq 20000$

A.13 Parameters in section Molecular Dynamics

- *Parameter name:* ATOMIC MASSES FILE

Value:

Default:

Description: [Standard] Input atomic masses file name. File format: atomicNumber1 atomicMass1 (row1), atomicNumber2 atomicMass2 (row2) and so on. Units: a.m.u.

Possible values: Any string

- *Parameter name:* BOMD

Value: false

Default: false

Description: [Standard] Perform Born-Oppenheimer molecular dynamics.

Possible values: A boolean value (true or false)

- *Parameter name:* EXTRAPOLATE DENSITY

Value: 0

Default: 0

Description: [Standard] Parameter controlling the reuse of ground-state density during molecular dynamics. The options are 0 default setting where superposition of atomic densities is the initial rho, 1 (second order extrapolation of density), and 2 (extrapolation of split density and the atomic densities are added) Option 2 is not enabled for spin-polarized case. Default setting is 0.

Possible values: An integer n such that $0 \leq n \leq 2$

- *Parameter name:* MAX JACOBIAN RATIO FACTOR

Value: 1.5

Default: 1.5

Description: [Developer] Maximum scaling factor for maximum jacobian ratio of FEM mesh when mesh is deformed.

Possible values: A floating point number v such that $0.9 \leq v \leq 3$

- *Parameter name:* MAX WALL TIME

Value: 2592000.0

Default: 2592000.0

Description: [Standard] Maximum Wall Time in seconds

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* NUMBER OF STEPS

Value: 1000

Default: 1000

Description: [Standard] Number of time steps.

Possible values: An integer n such that $0 \leq n \leq 200000$

- Parameter name:* **STARTING TEMPERATURE**

Value: 300.0

Default: 300.0

Description: [Standard] Starting temperature in K for MD simulation.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* **TEMPERATURE CONTROLLER TYPE**

Value: NO_CONTROL

Default: NO_CONTROL

Description: [Standard] Method of controlling temperature in the MD run. NO_CONTROL is the default option.

Possible values: Any one of NO_CONTROL, RESCALE, NOSE_HOVER_CHAINS, CSVR
- Parameter name:* **THERMOSTAT TIME CONSTANT**

Value: 100

Default: 100

Description: [Standard] Ratio of Time constant of thermostat and MD timestep.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* **TIME STEP**

Value: 0.5

Default: 0.5

Description: [Standard] Time step in femtoseconds.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* **TRACKING ATOMIC NO**

Value: 0

Default: 0

Description: [Standard] The atom Number to track.

Possible values: An integer n such that $0 \leq n \leq 200000$

A.14 Parameters in section Parallelization

- Parameter name:* **BAND PARAL OPT**

Value: true

Default: true

Description: [Standard] Uses a more optimal route for band parallelization but at the cost of extra wavefunctions memory.

Possible values: A boolean value (true or false)
- Parameter name:* **MPI ALLREDUCE BLOCK SIZE**

Value: 100.0

Default: 100.0

Description: [Advanced] Block message size in MB used to break a single MPI_Allreduce call on wavefunction vectors data into multiple MPI_Allreduce calls. This is useful on certain architectures

which take advantage of High Bandwidth Memory to improve efficiency of MPI operations. This variable is relevant only if NPBAND>1. Default value is 100.0 MB.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* NPBAND

Value: 1

Default: 1

Description: [Standard] Number of groups of MPI tasks across which the work load of the bands is parallelised. NPKPT times NPBAND must be a divisor of total number of MPI tasks. Further, NPBAND must be less than or equal to NUMBER OF KOHN-SHAM WAVEFUNCTIONS.

Possible values: An integer n such that $1 \leq n \leq 2147483647$

- *Parameter name:* NPKPT

Value: 1

Default: 1

Description: [Standard] Number of groups of MPI tasks across which the work load of the irreducible k-points is parallelised. NPKPT times NPBAND must be a divisor of total number of MPI tasks. Further, NPKPT must be less than or equal to the number of irreducible k-points.

Possible values: An integer n such that $1 \leq n \leq 2147483647$

A.15 Parameters in section Poisson problem parameters

- *Parameter name:* GPU MODE

Value: false

Default: false

Description: [Advanced] Toggle GPU MODE in Poisson solve.

Possible values: A boolean value (true or false)

- *Parameter name:* MAXIMUM ITERATIONS

Value: 20000

Default: 20000

Description: [Advanced] Maximum number of iterations to be allowed for Poisson problem convergence.

Possible values: An integer n such that $0 \leq n \leq 20000$

- *Parameter name:* TOLERANCE

Value: 1e-10

Default: 1e-10

Description: [Advanced] Absolute tolerance on the residual as stopping criterion for Poisson problem convergence.

Possible values: A floating point number v such that $0 \leq v \leq 1$

A.16 Parameters in section Post-processing Options

- *Parameter name:* READ ATOMIC WFC PDOS FROM PSP FILE

Value: false

Default: false

Description: [Standard] Read atomic wavefunctions from the pseudopotential file for computing projected density of states. When set to false atomic wavefunctions from the internal database are read, which correspond to sg15 ONCV pseudopotentials.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE BANDS

Value: false

Default: false

Description: [Standard] Write bands for every k-point to an outputfile called 'bands.out' in the units of Ha. This can be used after GS (Ground-state) or NSCF (Non-Self consistent field iteration) modes of solve. If it is set to true, Fermi energy is obtained from "fermiEnergy.out" file, created from previous GS calculation with "SAVE RHO DATA" set to true. Outputs a file name 'bands.out'. The first line has 2 entries with first one denoting the number of k-points and second entry denoting the number of eigenvalues(bands) for each k-point. Subsequent lines have 4 columns with first column indicating the k-point index, second column indicating band index, third column indicating corresponding eigenvalue and fourth column indicating the corresponding occupation number.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE DENSITY FE MESH

Value: false

Default: false

Description: [Standard] Writes DFT ground state electron-density solution fields (FEM mesh nodal values) to densityOutput.vtu file for visualization purposes. The electron-density solution field in densityOutput.vtu is named density. In case of spin-polarized calculation, two additional solution fields- density_0 and density_1 are also written where 0 and 1 denote the spin indices. In the case of geometry optimization, the electron-density corresponding to the last ground-state solve is written. Default: false.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE DENSITY OF STATES

Value: false

Default: false

Description: [Standard] Computes density of states using Lorentzians. Uses specified Temperature for SCF as the broadening parameter. Outputs a file name 'dosData.out' containing two columns with first column indicating the energy in eV relative to the Fermi energy and second column indicating the density of states. In case of collinear spin polarization, the second and third columns indicate the spin-up and spin-down density of states.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE DENSITY QUAD DATA

Value: false

Default: false

Description: [Standard] Writes DFT ground state electron-density solution fields at generally non-uniform quadrature points to a .txt file for post-processing. There will be six columns (in case of collinear spin polarization) and 5 columns in case of spin-restricted calculation. The first three columns are the quadrature point cartesian coordinates (non-uniform grid with origin at cell center), fourth column is the quadrature integration weight incorporating the determinant of FE cell jacobian, and the fifth and sixth columns are the spin-up and spin-down densities in case of collinear spin polarization. In case of spin-restricted calculation, the fifth column has the total density. Default: false.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE LOCALIZATION LENGTHS

Value: false

Default: false

Description: [Standard] Computes localization lengths of all wavefunctions which is defined as the deviation around the mean position of a given wavefunction. Outputs a file name 'localizationLengths.out' containing 2 columns with first column indicating the wavefunction index and second column indicating localization length of the corresponding wavefunction.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE LOCAL DENSITY OF STATES

Value: false

Default: false

Description: [Standard] Computes local density of states on each atom using Lorentzians. Uses specified Temperature for SCF as the broadening parameter. Outputs a file name 'ldosData.out' containing NUMATOM+1 columns with first column indicating the energy in eV and all other NUMATOM columns indicating local density of states for each of the NUMATOM atoms.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE PROJECTED DENSITY OF STATES

Value: false

Default: false

Description: [Standard] Computes projected density of states on each atom using Lorentzians. Uses specified Temperature for SCF as the broadening parameter. Outputs a file name 'pdosData_x' with x denoting atomID. This file contains columns with first column indicating the energy in eV and all other columns indicating projected density of states corresponding to single atom wavefunctions.

Possible values: A boolean value (true or false)

- *Parameter name:* WRITE WFC FE MESH

Value: false

Default: false

Description: [Standard] Writes DFT ground state wavefunction solution fields (FEM mesh nodal values) to wfcOutput.vtu file for visualization purposes. The wavefunction solution fields in wfcOutput.vtu are named wfc_s_k_i in case of spin-polarized calculations and wfc_k_i otherwise, where s denotes the spin index (0 or 1), k denotes the k point index starting from 0, and i denotes the Kohn-Sham wavefunction index starting from 0. In the case of geometry optimization, the wavefunctions corresponding to the last ground-state solve are written. Default: false.

Possible values: A boolean value (true or false)

A.17 Parameters in section SCF Checkpointing and Restart

- *Parameter name:* LOAD RHO DATA
Value: false
Default: false
Description: [Standard] Loads charge density and mesh triangulation data from file.
Possible values: A boolean value (true or false)
- *Parameter name:* RESTART SP FROM NO SP
Value: false
Default: false
Description: [Standard] Enables ground-state solve for SPIN POLARIZED case reading the SPIN UNPOLARIZED density from the checkpoint files, and use the START MAGNETIZATION to compute the spin up and spin down densities. This option is only valid for CHK TYPE=2 and RESTART FROM CHK=true. Default false..
Possible values: A boolean value (true or false)
- *Parameter name:* SAVE RHO DATA
Value: false
Default: false
Description: [Standard] Saves charge density and mesh triangulation data for restart, if SOLVER MODE is GS then the save is done every 10 scf iterations, otherwise it is done after each converged scf solve. If the value is "true", the SOLVER MODE is GS and if the SCF loop converges, an outputfile "fermiEnergy.out" is written that contains the fermi energy in the units of Ha. This Fermi energy is used when "WRITE BANDS" is true.
Possible values: A boolean value (true or false)

A.18 Parameters in section SCF parameters

- *Parameter name:* ADAPT ANDERSON MIXING PARAMETER
Value: false
Default: false
Description: [Standard] Boolean parameter specifying whether to adapt the Anderson mixing parameter based on algorithm 1 in [CPC. 292, 108865 (2023)].
Possible values: A boolean value (true or false)
- *Parameter name:* COMPUTE ENERGY EACH ITER
Value: false
Default: false
Description: [Advanced] Boolean parameter specifying whether to compute the total energy at the end of every SCF. Setting it to false can lead to some computational time savings. Default value is false but is internally set to true if VERBOSITY==5
Possible values: A boolean value (true or false)

- Parameter name:* USE ENERGY RESIDUAL METRIC

Value: false

Default: false

Description: [Advanced] Boolean parameter specifying whether to use the energy residual metric (equation 7.23 of Richard Matrin second edition) for convergence check. Setting it to false can lead to some computational time savings. Default value is false

Possible values: A boolean value (true or false)
- Parameter name:* CONSTRAINT MAGNETIZATION

Value: false

Default: false

Description: [Standard] Boolean parameter specifying whether to keep the starting magnetization fixed through the SCF iterations. Default is FALSE

Possible values: A boolean value (true or false)
- Parameter name:* KERKER MIXING PARAMETER

Value: 0.05

Default: 0.05

Description: [Standard] Mixing parameter to be used in Kerker mixing scheme which usually represents Thomas Fermi wavevector (k_{TF}^2).

Possible values: A floating point number v such that $0 \leq v \leq 1000$
- Parameter name:* MAXIMUM ITERATIONS

Value: 200

Default: 200

Description: [Standard] Maximum number of iterations to be allowed for SCF convergence

Possible values: An integer n such that $1 \leq n \leq 1000$
- Parameter name:* MIXING HISTORY

Value: 10

Default: 10

Description: [Standard] Number of SCF iteration history to be considered for density mixing schemes. For metallic systems, a mixing history larger than the default value provides better scf convergence.

Possible values: An integer n such that $1 \leq n \leq 1000$
- Parameter name:* MIXING METHOD

Value: ANDERSON

Default: ANDERSON

Description: [Standard] Method for density mixing. ANDERSON is the default option.

Possible values: Any one of ANDERSON, ANDERSON_WITH_KERKER, ANDERSON_WITH_RESTA, LOW_RANK_DIELECM_PRECOND
- Parameter name:* MIXING PARAMETER

Value: 0.0

Default: 0.0

Description: [Standard] Mixing parameter to be used in density mixing schemes. For default value of 0.0, it is heuristically set for different mixing schemes (0.2 for Anderson, and 0.5 for Kerker and LRD).

Possible values: A floating point number v such that $-1e-12 \leq v \leq 1$

- *Parameter name:* SPIN MIXING ENHANCEMENT FACTOR

Value: 4.0

Default: 4.0

Description: [Standard] Scales the mixing parameter for the spin densities as SPIN MIXING ENHANCEMENT FACTOR times MIXING PARAMETER. This parameter is not used for LOW_RANK_DIELECM_PFR mixing method.

Possible values: A floating point number v such that $-1e-12 \leq v \leq 10$

- *Parameter name:* RESTA FERMI WAVEVECTOR

Value: 5.81

Default: 5.81

Description: [Standard] Fermi wavevector estimate (in Bohr⁻¹) to be used in the Resta preconditioner.

Possible values: A floating point number v such that $0 \leq v \leq 1000$

- *Parameter name:* RESTA SCREENING LENGTH

Value: 6.61

Default: 6.61

Description: [Standard] Screening length estimate (in Bohr) to be used in the Resta preconditioner.

Possible values: A floating point number v such that $0 \leq v \leq 1000$

- *Parameter name:* STARTING WFC

Value: RANDOM

Default: RANDOM

Description: [Standard] Sets the type of the starting Kohn-Sham wavefunctions guess: Atomic(Superposition of single atom atomic orbitals. Atom types for which atomic orbitals are not available, random wavefunctions are taken. Currently, atomic orbitals data is not available for all atoms.), Random(The starting guess for all wavefunctions are taken to be random). Default: RANDOM.

Possible values: Any one of ATOMIC, RANDOM

- *Parameter name:* TEMPERATURE

Value: 500.0

Default: 500.0

Description: [Standard] Fermi-Dirac smearing temperature (in Kelvin).

Possible values: A floating point number v such that $1e-05 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* TOLERANCE

Value: 1e-05

Default: 1e-05

Description: [Standard] SCF iterations stopping tolerance in terms of L_2 norm of the electron-density difference between two successive iterations. The default tolerance of is set to a tight value of 1e-5 for accurate ionic forces and cell stresses keeping structural optimization and molecular dynamics in mind. A tolerance of 1e-4 would be accurate enough for calculations without structural optimization

and dynamics. CAUTION: A tolerance close to $1e-7$ or lower can deteriorate the SCF convergence due to the round-off error accumulation.

Possible values: A floating point number v such that $1e-12 \leq v \leq 1$

A.19 Parameters in section SCF parameters/Eigen-solver parameters

- *Parameter name:* ALGO

Value: NORMAL

Default: NORMAL

Description: [Standard] In the FAST mode, spectrum splitting technique is used in Rayleigh-Ritz step, and mixed precision arithmetic algorithms are used in Rayleigh-Ritz and Cholesky factorization based orthogonalization step. For spectrum splitting, 85 percent of the total number of wavefunctions are taken to be core states, which holds good for most systems including metallic systems assuming NUMBER OF KOHN-SHAM WAVEFUNCTIONS to be around 10 percent more than $N/2$. FAST setting is strongly recommended for large-scale ($> 10k$ electrons) system sizes. Both NORMAL and FAST setting use Chebyshev filtered subspace iteration technique. If manual options for mixed precision and spectrum splitting are being used, please use NORMAL setting for ALGO. Default setting is NORMAL.

Possible values: Any one of NORMAL, FAST

- *Parameter name:* ALLOW MULTIPLE PASSES POST FIRST SCF

Value: true

Default: true

Description: [Advanced] Allow multiple chebyshev filtering passes in the SCF iterations after the first one. Default setting is true.

Possible values: A boolean value (true or false)

- *Parameter name:* CHEBYSHEV FILTER TOLERANCE

Value: 0.0

Default: 0.0

Description: [Advanced] Parameter specifying the accuracy of the occupied eigenvectors close to the Fermi-energy computed using Chebyshev filtering subspace iteration procedure. For default value of 0.0, we heuristically set the value between $1e-3$ and $5e-2$ depending on the MIXING METHOD used.

Possible values: A floating point number v such that $-1e-12 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* CHEBYSHEV POLYNOMIAL DEGREE

Value: 0

Default: 0

Description: [Advanced] Chebyshev polynomial degree to be employed for the Chebyshev filtering subspace iteration procedure to dampen the unwanted spectrum of the Kohn-Sham Hamiltonian. If set to 0, a default value depending on the upper bound of the eigen-spectrum is used. See Phani Motamarri et.al., J. Comp. Phys. 253, 308-343 (2013).

Possible values: An integer n such that $0 \leq n \leq 2000$

- *Parameter name:* CHEBYSHEV POLYNOMIAL DEGREE SCALING FACTOR FIRST SCF

Value: 1.34

Default: 1.34

Description: [Advanced] Chebyshev polynomial degree first scf scaling factor. Only activated for pseudopotential calculations.

Possible values: A floating point number v such that $0 \leq v \leq 2000$

- *Parameter name:* CHEBY WFC BLOCK SIZE

Value: 200

Default: 200

Description: [Advanced] Chebyshev filtering procedure involves the matrix-matrix multiplication where one matrix corresponds to the discretized Hamiltonian and the other matrix corresponds to the wavefunction matrix. The matrix-matrix multiplication is accomplished in a loop over the number of blocks of the wavefunction matrix to reduce the memory footprint of the code. This parameter specifies the block size of the wavefunction matrix to be used in the matrix-matrix multiplication. The optimum value is dependent on the computing architecture. For optimum work sharing during band parallelization ($\text{NPBAND} > 1$), we recommend adjusting CHEBY WFC BLOCK SIZE and NUMBER OF KOHN-SHAM WAVEFUNCTIONS such that $\text{NUMBER OF KOHN-SHAM WAVEFUNCTIONS} / \text{NPBAND} / \text{CHEBY WFC BLOCK SIZE}$ equals an integer value. Default value is 200.

Possible values: An integer n such that $1 \leq n \leq 2147483647$

- *Parameter name:* HIGHEST STATE OF INTEREST FOR CHEBYSHEV FILTERING

Value: 0

Default: 0

Description: [Standard] The highest state till which the Kohn Sham wavefunctions are computed accurately during Chebyshev filtering in a NSCF calculation. By default, this is set to the state corresponding to Fermi energy. It is strongly encouraged to have 10-15 percent buffer between this parameter and the total number of wavefunctions employed for the SCF calculation

Possible values: An integer n such that $0 \leq n \leq 2147483647$

- *Parameter name:* NUMBER OF KOHN-SHAM WAVEFUNCTIONS

Value: 0

Default: 0

Description: [Standard] Number of Kohn-Sham wavefunctions to be computed. For spin-polarized calculations, this parameter denotes the number of Kohn-Sham wavefunctions to be computed for each spin. A recommended value for this parameter is to set it to $N/2 + N_b$ where N is the number of electrons. Use N_b to be 5-10 percent of $N/2$ for insulators and for metals use N_b to be 10-20 percent of $N/2$. If 5-20 percent of $N/2$ is less than 10 wavefunctions, set N_b to be atleast 10. Default value of 0 automatically sets the number of Kohn-Sham wavefunctions close to 20 percent more than $N/2$. CAUTION: use more states when using higher electronic temperature.

Possible values: An integer n such that $0 \leq n \leq 2147483647$

- *Parameter name:* ORTHOGONALIZATION TYPE

Value: Auto

Default: Auto

Description: [Advanced] Parameter specifying the type of orthogonalization to be used: GS(Gram-Schmidt Orthogonalization using SLEPc library) and CGS(Cholesky-Gram-Schmidt Orthogonalization). Auto is the default and recommended option, which chooses GS for all-electron case and CGS for pseudopotential case. On GPUs CGS is the only route currently implemented.

Possible values: Any one of GS, CGS, Auto

- Parameter name:* OVERLAP COMPUTE COMMUN CHEBY

Value: true

Default: true

Description: [Advanced] Overlap communication and computation in Chebyshev filtering. This option can only be activated for USE GPU=true. Default setting is true.

Possible values: A boolean value (true or false)
- Parameter name:* OVERLAP COMPUTE COMMUN ORTHO RR

Value: true

Default: true

Description: [Advanced] Overlap communication and computation in orthogonalization and Rayleigh-Ritz. This option can only be activated for USE GPU=true. Default setting is true.

Possible values: A boolean value (true or false)
- Parameter name:* RESTRICT TO SINGLE FILTER PASS

Value: false

Default: false

Description: [Advanced] Restrict to single chebyshev filter pass in each SCF. This setting is only used for timing measurements of stable single SCF iteration.

Possible values: A boolean value (true or false)
- Parameter name:* REUSE LANCZOS UPPER BOUND

Value: false

Default: false

Description: [Advanced] Reuse upper bound of unwanted spectrum computed in the first SCF iteration via Lanczos iterations. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* SCALAPACKPROCS

Value: 0

Default: 0

Description: [Advanced] Uses a processor grid of SCALAPACKPROCS times SCALAPACKPROCS for parallel distribution of the subspace projected matrix in the Rayleigh-Ritz step and the overlap matrix in the Cholesky-Gram-Schmidt step. Default value is 0 for which a thumb rule is used (see <http://netlib.org/scalapack/slug/node106.html>). If ELPA is used, twice the value obtained from the thumb rule is used as ELPA scales much better than ScaLAPACK.

Possible values: An integer n such that $0 \leq n \leq 300$
- Parameter name:* SCALAPACK BLOCK SIZE

Value: 0

Default: 0

Description: [Advanced] ScaLAPACK process grid block size. Also sets the block size for ELPA if linked to ELPA. Default value of zero sets a heuristic block size. Note that if ELPA GPU KERNEL is set to true and ELPA is configured to run on GPUs, the SCALAPACK BLOCK SIZE is set to a power of 2.

Possible values: An integer n such that $0 \leq n \leq 300$

- Parameter name:* SPECTRUM SPLIT CORE EIGENSTATES

Value: 0

Default: 0

Description: [Advanced] Number of lowest Kohn-Sham eigenstates which should not be included in the Rayleigh-Ritz diagonalization. In other words, only the eigenvalues and eigenvectors corresponding to the higher eigenstates (Number of Kohn-Sham wavefunctions minus the specified core eigenstates) are computed in the diagonalization of the projected Hamiltonian. This value is usually chosen to be the sum of the number of core eigenstates for each atom type multiplied by number of atoms of that type. This setting is recommended for large systems (greater than 5000 electrons). Default value is 0 i.e., no core eigenstates are excluded from the Rayleigh-Ritz projection step.

Possible values: An integer n such that $0 \leq n \leq 2147483647$
- Parameter name:* SPECTRUM SPLIT STARTING SCF ITER

Value: 0

Default: 0

Description: [Advanced] SCF iteration no beyond which spectrum splitting based can be used.

Possible values: An integer n such that $0 \leq n \leq 2147483647$
- Parameter name:* SUBSPACE PROJ SHEP GPU

Value: true

Default: true

Description: [Advanced] Solve a standard hermitian eigenvalue problem in the Rayleigh Ritz step instead of a generalized hermitian eigenvalue problem on GPUs. Default setting is true.

Possible values: A boolean value (true or false)
- Parameter name:* SUBSPACE ROT DOFS BLOCK SIZE

Value: 10000

Default: 10000

Description: [Developer] This block size is used for memory optimization purposes in subspace rotation step in Cholesky-Gram-Schmidt orthogonalization and Rayleigh-Ritz steps. Default value is 10000.

Possible values: An integer n such that $1 \leq n \leq 2147483647$
- Parameter name:* USE ELPA

Value: true

Default: true

Description: [Standard] Use ELPA instead of ScaLAPACK for diagonalization of subspace projected Hamiltonian and Cholesky-Gram-Schmidt orthogonalization. Default setting is true.

Possible values: A boolean value (true or false)
- Parameter name:* USE MIXED PREC CGS 0

Value: false

Default: false

Description: [Advanced] Use mixed precision arithmetic in overlap matrix computation step of CGS orthogonalization, if ORTHOGONALIZATION TYPE is set to CGS. Default setting is false.

Possible values: A boolean value (true or false)

- Parameter name:* USE MIXED PREC CGS SR

Value: false

Default: false

Description: [Advanced] Use mixed precision arithmetic in subspace rotation step of CGS orthogonalization, if ORTHOGONALIZATION TYPE is set to CGS. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* USE SINGLE PREC COMMUN CHEBY

Value: false

Default: false

Description: [Advanced] Use single precision communication in Chebyshev filtering. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* USE MIXED PREC COMMUN ONLY XTX XTHX

Value: false

Default: false

Description: [Advanced] Use mixed precision communication only for XtX and XtHX instead of mixed precision compute and communication. This setting has been found to be more optimal on certain architectures. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* USE SINGLE PREC CHEBY

Value: false

Default: false

Description: [Advanced] Use a modified single precision algorithm for Chebyshev filtering. This cannot be used in conjunction with spectrum splitting. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* USE MIXED PREC RR_SR

Value: false

Default: false

Description: [Advanced] Use mixed precision arithmetic in Rayleigh-Ritz subspace rotation step. Default setting is false.

Possible values: A boolean value (true or false)
- Parameter name:* USE MIXED PREC XTHX SPECTRUM SPLIT

Value: false

Default: false

Description: [Advanced] Use mixed precision arithmetic in computing subspace projected Kohn-Sham Hamiltonian when SPECTRUM SPLIT CORE EIGENSTATES>0. Default setting is false.

Possible values: A boolean value (true or false)

- *Parameter name:* WFC BLOCK SIZE

Value: 400

Default: 400

Description: [Advanced] This parameter specifies the block size of the wavefunction matrix to be used for memory optimization purposes in the orthogonalization, Rayleigh-Ritz, and density computation steps. The optimum block size is dependent on the computing architecture. For optimum work sharing during band parallelization (NPBAND > 1), we recommend adjusting WFC BLOCK SIZE and NUMBER OF KOHN-SHAM WAVEFUNCTIONS such that NUMBER OF KOHN-SHAM WAVEFUNCTIONS/NPBAND/WFC BLOCK SIZE equals an integer value. Default value is 400.

Possible values: An integer n such that $1 \leq n \leq 2147483647$

- *Parameter name:* XTHX CORE EIGENSTATES

Value: 0

Default: 0

Description: [Advanced] For mixed precision optimization.

Possible values: An integer n such that $0 \leq n \leq 2147483647$

A.20 Parameters in section SCF parameters/LOW RANK DIELECM PRECOND

- *Parameter name:* ADAPTIVE RANK REL TOL

Value: 0.3

Default: 0.3

Description: [Standard] Relative error metric on the low rank approximation error that adaptively sets the rank in each SCF iteration step.

Possible values: A floating point number v such that $0 \leq v \leq 1$

- *Parameter name:* BETA TOL

Value: 0.1

Default: 0.1

Description: [Advanced] Sets tolerance on deviation of linear indicator value from the ideal value of 1.0. For METHOD SUB TYPE=ACCUMULATED_ADAPTIVE.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$

- *Parameter name:* ESTIMATE JAC CONDITION NO

Value: false

Default: false

Description: [Advanced] Estimate condition number of the Jacobian at the final SCF iteration step using a low rank approximation with ADAPTIVE RANK REL TOL=1.0e-5.

Possible values: A boolean value (true or false)

- *Parameter name:* METHOD SUB TYPE

Value: ADAPTIVE

Default: ADAPTIVE

Description: [Advanced] Method subtype for LOW_RANK_DIELECM_PRECOND.

Possible values: Any one of ADAPTIVE, ACCUMULATED_ADAPTIVE

- Parameter name:* POISSON SOLVER ABS TOL

Value: 1e-6

Default: 1e-6

Description: [Advanced] Absolute poisson solver tolerance for electrostatic potential response computation.

Possible values: A floating point number v such that $0 \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* STARTING NORM LARGE DAMPING

Value: 2.0

Default: 2.0

Description: [Advanced] L2 norm electron density difference below which damping parameter is set to SCF parameters::MIXING PARAMETER, otherwise set to 0.1.

Possible values: A floating point number v such that $0 \leq v \leq 10$
- Parameter name:* USE SINGLE PREC DENSITY RESPONSE

Value: false

Default: false

Description: [Advanced] Turns on single precision optimization in density response computation.

Possible values: A boolean value (true or false)

A.21 Parameters in section NEB

- Parameter name:* ALLOW IMAGE FREEZING

Value: false

Default: false

Description: If true images less than threshold will freeze for optimization step

Possible values: A boolean value (true or false)
- Parameter name:* BFGS STEP METHOD

Value: QN

Default: QN

Description: [Standard] Method for computing update step in BFGS. Quasi-Newton step (default) or Rational Function Step as described in JPC 1985, 89:52-57.

Possible values: Any one of QN, RFO
- Parameter name:* ION RELAX FLAGS FILE

Value:

Default:

Description: [Standard] File specifying the permission flags (1-free to move, 0-fixed) and external forces for the 3-coordinate directions and for all atoms. File format (example for two atoms with atom 1 fixed and atom 2 free and 0.01 Ha/Bohr force acting on atom 2): 0 0 0 0.0 0.0 0.0(row1), 1 1 1 0.0 0.0 0.0(row2). External forces are optional.

Possible values: Any string

- Parameter name:* LBFGS HISTORY

Value: 5

Default: 5

Description: [Standard] Number of previous steps to considered for the LBFGS update.

Possible values: An integer n such that $1 \leq n \leq 20$
- Parameter name:* MAXIMUM ION UPDATE STEP

Value: 0.5

Default: 0.5

Description: [Standard] Sets the maximum allowed step size (displacement in a.u.) during ion relaxation.

Possible values: A floating point number v such that $0 \leq v \leq 5$
- Parameter name:* MAXIMUM NUMBER OF NEB ITERATIONS

Value: 100

Default: 100

Description: [Standard] Maximum number of NEB iterations that will be performed in the simulation

Possible values: An integer n such that $1 \leq n \leq 250$
- Parameter name:* MAXIMUM SPRING CONSTANT

Value: 5e-3

Default: 5e-3

Description: [Standard] Sets the maximum allowable spring constant in (Ha/bohr²)

Possible values: A floating point number v such that $-\text{MAX_DOUBLE} \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* MAX LINE SEARCH ITER

Value: 5

Default: 5

Description: [Standard] Sets the maximum number of line search iterations in the case of CGPRP. Default is 5.

Possible values: An integer n such that $1 \leq n \leq 100$
- Parameter name:* MINIMUM SPRING CONSTANT

Value: 2e-3

Default: 2e-3

Description: [Standard] Sets the minimum allowable spring constant in (Ha/bohr²)

Possible values: A floating point number v such that $-\text{MAX_DOUBLE} \leq v \leq \text{MAX_DOUBLE}$
- Parameter name:* NEB COORDINATES FILE

Value:

Default:

Description: [Standard] Atomic-coordinates input file name. For fully non-periodic domain give Cartesian coordinates of the atoms (in a.u) with respect to origin at the center of the domain. For periodic and semi-periodic domain give fractional coordinates of atoms. File format (example for two atoms):

Atom1-atomic-charge Atom1-valence-charge x1 y1 z1 (row1), Atom2-atomic-charge Atom2-valence-charge x2 y2 z2 (row2). The number of rows must be equal to NATOMS, and number of unique atoms must be equal to NATOM TYPES.

Possible values: Any string

- *Parameter name:* NEB DOMAIN VECTORS FILE

Value:

Default:

Description: [Standard] Atomic-coordinates input file name. For fully non-periodic domain give Cartesian coordinates of the atoms (in a.u) with respect to origin at the center of the domain. For periodic and semi-periodic domain give fractional coordinates of atoms. File format (example for two atoms): Atom1-atomic-charge Atom1-valence-charge x1 y1 z1 (row1), Atom2-atomic-charge Atom2-valence-charge x2 y2 z2 (row2). The number of rows must be equal to NATOMS, and number of unique atoms must be equal to NATOM TYPES.

Possible values: Any string

- *Parameter name:* NEB OPT SOLVER

Value: LBFGS

Default: LBFGS

Description: [Standard] Method for Ion relaxation solver. LBFGS is the default

Possible values: Any one of BFGS, LBFGS, CGPRP

- *Parameter name:* NUMBER OF IMAGES

Value: 7

Default: 7

Description: [Standard] NUMBER OF IMAGES:Default option is 7. When NEB is triggered this controls the total number of images along the MEP including the end points

Possible values: An integer n such that $1 \leq n \leq 50$

- *Parameter name:* PATH THRESHOLD

Value: 5e-4

Default: 5e-4

Description: [Standard] Simulation stops when the error(norm of force orthogonal to path in Ha/bohr) is less than PATH THRESHOLD

Possible values: A floating point number v such that $-\text{MAX_DOUBLE} \leq v \leq \text{MAX_DOUBLE}$

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